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Paleomagnetism and $^{40}Ar/^{39}Ar$ geochronology of the high-grade metamorphic rocks of the Jequié block, São Francisco Craton: Atlantica, Ur and beyond

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ABSTRACT

Paleomagnetic, magnetic anisotropy data and $^{40}Ar/^{39}Ar$ ages are presented for high-grade metamorphic rocks of the Jequié block, São Francisco Craton. Jequié charnockites and enderbites gneisses from the eastern border of the block present northern and steep-downward directions, carried by Ti-poor titanomagnetite with high unblocking temperatures (550–600 °C). A mean direction for 12 sites of enderbites from the eastern sector yielded a magnetic component A (D_m = 47.0°, I_m = 75.7°, α_{95} = 6.3°, K = 48.8) with a corresponding paleomagnetic pole at 339.6°E, 5.4°N (A_{95} = 11.2°). Sites sampled on other metamorphic rocks including granulite-facies, tonalites and dacites yield different magnetic components. Anisotropy of magnetic susceptibility measured for all sampling sites shows a high degree of anisotropy (P = 1.121–1.881), with NE- or NW-trending magnetic lineations and vertical magnetic foliations. These data were used to correct the mean site directions for all components. While the other components presented larger scatter after anisotropy correction, the component A shows a slightly tighter clustering of magnetic directions (D_m = 61.2°, I_m = 76.5°, α_{95} = 5.4°, K = 66.2) giving a new, anisotropy corrected paleomagnetic pole at 342.1°E, −0.5°N (A_{95} = 9.6°). ^{40}Ar–^{39}Ar plateau ages of 2035 ± 4 Ma (hornblende) and 1876 ± 4 Ma (biotite) obtained for one of the samples with component A imply a very low cooling-rate of 1.4 °C/Ma for these rocks. Based on these ages and corrected unblocking temperatures of the magnetic component A, we argue that the characteristic magnetization of the Jequié metamorphic rocks was acquired by high temperature thermo-chemical processes during regional cooling of the adjacent Itabuna–Salvador–Curaçá belt between 2089 and 1985 Ma (pole age likely at 1.99 Ga). Comparison of this pole with available paleomagnetic poles from South America and Africa suggests that neither Atlantica nor Ur have ever existed, and dispersed continental fragments dominated the paleogeography at ca. 2.0 Ga ago.

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1. Introduction

Reconstruction of the supercontinent cycles is important for understanding the dynamics of the Earth's tectonic history, as the assembly of supercontinents produces collisional mountain belts and sutures among other features, such as large igneous provinces and mafic dike swarms. At least three supercontinents have been proposed in the literature: a late Paleo- to Mesoproterozoic assembly known as Columbia, the Mesoproterozoic Rodinia and the late Paleozoic to Mesozoic supercontinent Pangea. In the original model for Columbia (Rogers and Santosh, 2002), Paleoproterozoic belts formed the sutures between three main continental blocks: Atlantica (Amazonian, West Congo-São Francisco, Rio de la Plata and West Africa proto-cratons), Ur (Antarctica, Australia, India, Madagascar, Zimbabwe and Kaapvaal cratons) and Nena (North America, Siberia and Greenland, together with East Antarctica and Baltica). Recently, new paleogeographic models for Columbia (e.g., Zhao et al., 2002, 2004; Kusky et al., 2007; Hou et al., 2008; Bispo-Santos et al., 2008) suggest an emerging consensus for the assembly of Nena. However, the configurations of Atlantica and Ur remain a matter of contention. As usual paleomagnetic data coupled with reliable ages are essential to test these models, since they represent the method that provides ancient paleolatitude and paleo-orientation of continents through time.

This study reports new paleomagnetic and magnetic anisotropy data from high-grade metamorphic rocks belonging to the Jequié block one of the tectonic segments that makes up the Archean/Paleoproterozoic framework of the São Francisco Craton

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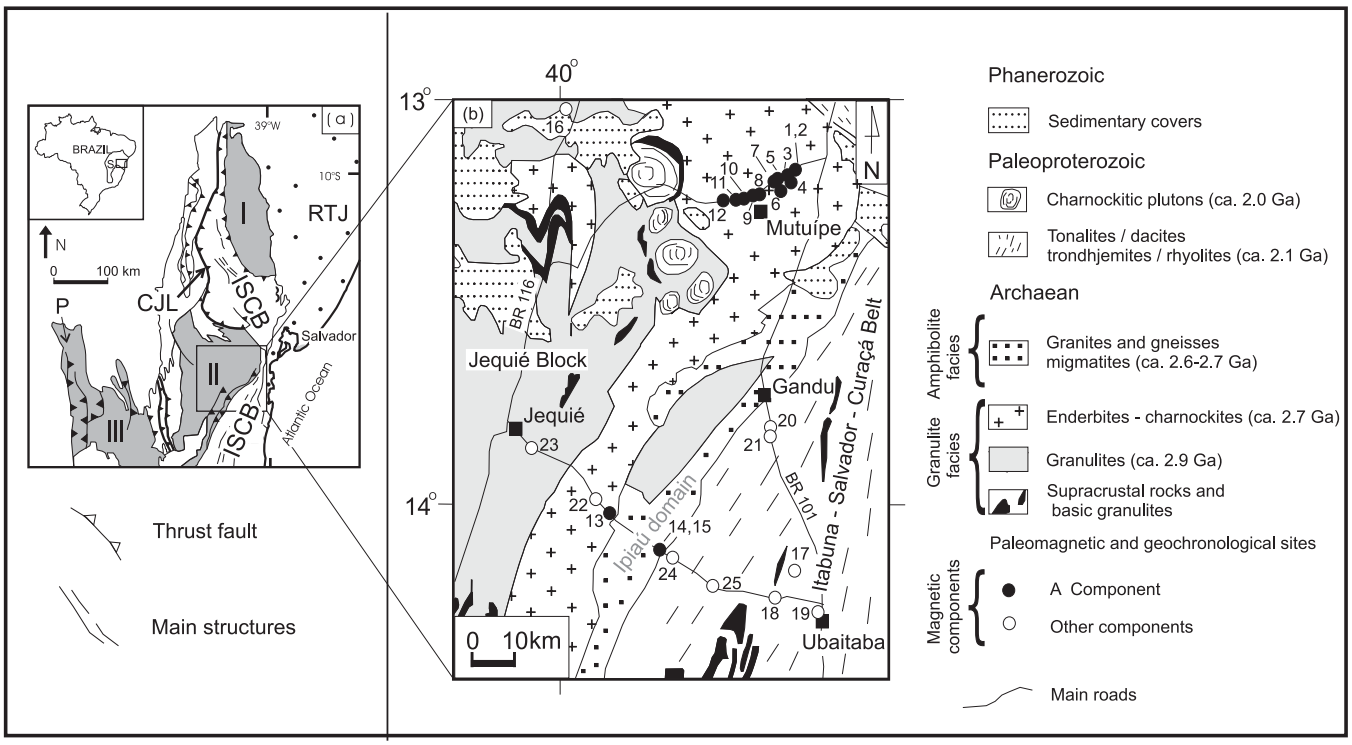


Fig. 1. (a) Geotectonic sketch of the northeastern sector of the São Francisco craton (after Barbosa and Sabaté, 2004). Archean blocks—I, Serrinha; II, Jequié; III, Gavião, P, Paramirim Province. Paleoproterozoic mobile belt: ISCB, Itabuna–Salvador–Curaçá belt; CJL, Contendas-Jacobina lineament; RTJ, Cretaceous Recôncavo-Tucano basin. Inset figure: S, São Francisco craton. (b) Simplified geological map of the studied area (after Teixeira et al., 2000; Barbosa and Sabaté, 2004). Paleomagnetic sampling sites – numbers from 1 to 25. Magnetic component isolated at each site is also shown.

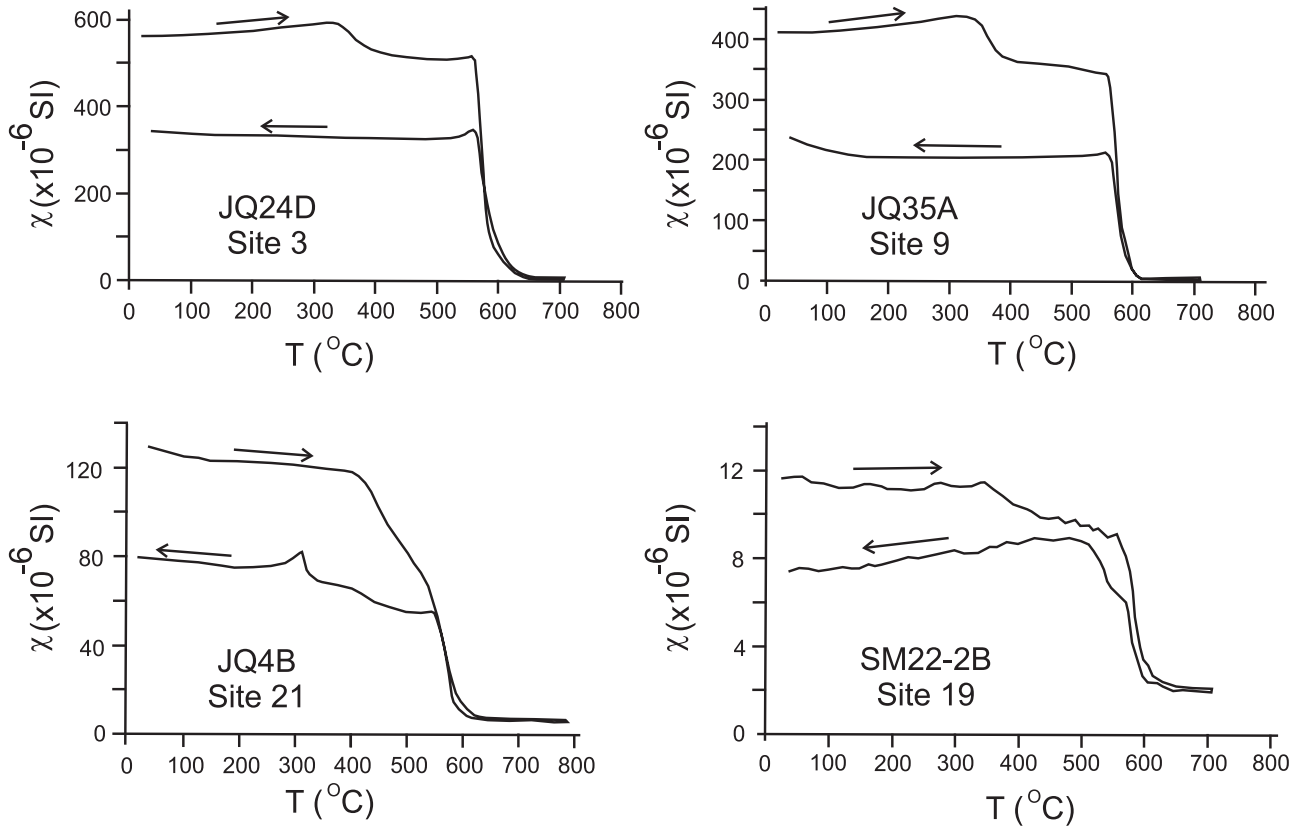


Fig. 2. Typical thermomagnetic curves (susceptibility versus temperature) performed in air. Arrows indicate heating and cooling cycles.

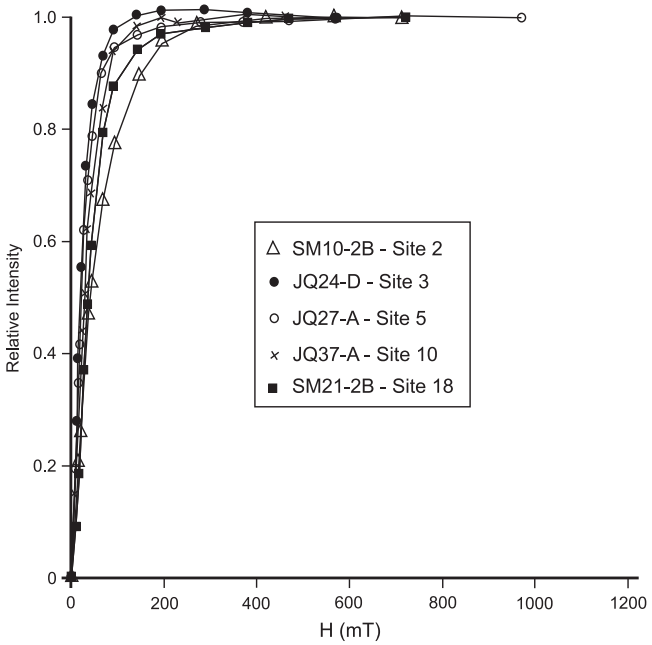


Fig. 3. Examples of isothermal remanent magnetization (IRM–relative intensity versus magnetic field) curves.

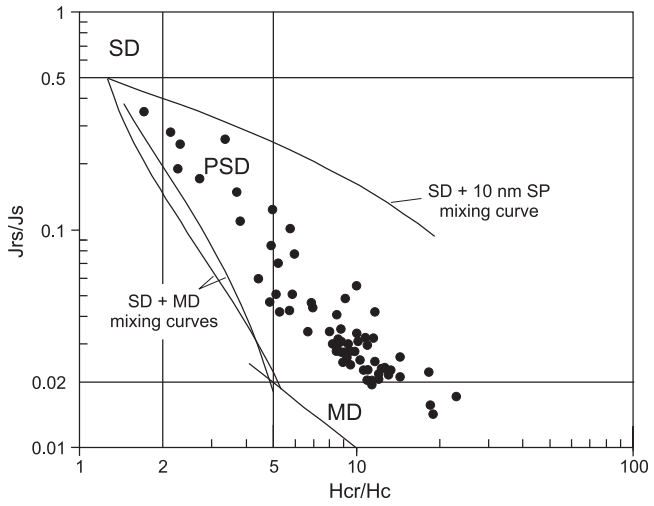


Fig. 5. Day's diagram: Jrs/Js versus Hcr/Hc ratios (Day et al., 1977). The limits between SD, PSD and MD fields are those proposed by Dunlop (2002a,b). Most hysteresis parameters are compatible with SD plus MD mixing curves as proposed by Dunlop (2002a,b).

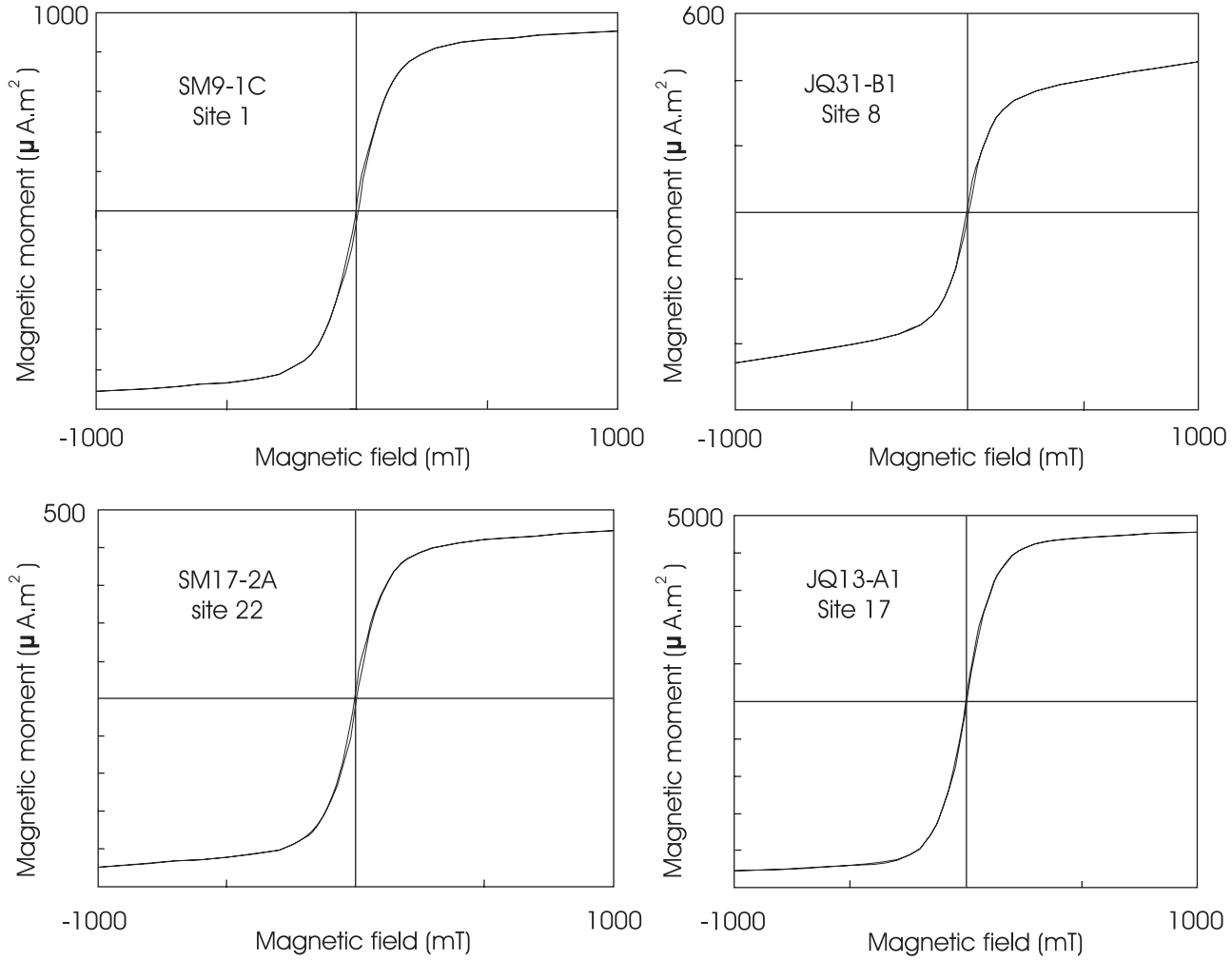


Fig. 4. Examples of hysteresis curves.

Table 1
Jequié paleomagnetic data for component A.

Site	Samples	Localization	*N*/*n*	Site mean directions						VGPs			
				Uncorrected		AMS corrected						AMS Corrected	
		Lat (°S)/Long (°W)		Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	α_{95} (°)	K	Plong. (°E)	Plat. (°N)	Plong. (°E)	Plat. (°N)
1	SM91-93	13.18/39.42	10/11	69.5	66.6	63.6	72.2	9.7	25.6	358.5	2.9	349.6	2.4
2	SM101-103	13.18/39.42	11/12	48.0	71.9	40.8	77.1	4.8	90.1	344.9	9.5	336.5	5.7
3	JQ23-24	13.18/39.43	8/8	340.0	78.2	74.4	74.1	4.3	163.9	312.9	8.2	349.1	−3.9
4	JQ21A-D	13.20/39.43	5/7	46.8	60.9	44.7	68.1	5.0	100.7	355.8	20.1	347.7	14.8
5	JQ25-27	13.18/39.47	9/9	37.1	75.3	22.5	81.8	3.1	283.5	337.0	9.2	326.6	1.7
6	JQ28-30	13.20/39.47	10/10	34.0	75.0	42.4	84.7	3.1	235.6	336.1	10.4	327.6	−5.4
7	SM111-113	13.23/39.50	12/12	42.2	65.2	56.8	63.3	4.3	104.2	349.3	18.8	357.9	12.5
8	JQ31A-C	13.23/39.53	5/5	118.7	74.3	119.5	72.8	11.9	42.3	348.9	−25.4	351.3	−26.5
9	JQ32-35	13.23/39.55	6/7	31.7	71.7	44.5	72.3	5.2	164.8	338.0	15.4	343.0	10.4
10	JQ36-38	13.25/39.58	9/9	48.0	73.7	66.5	69.5	3.1	274.7	342.7	7.6	353.8	2.8
11	JQ39-41	13.25/39.60	5/9	340.0	78.6	8.2	81.4	9.6	64.6	313.0	7.5	322.8	3.4
12	SM121-123	13.27/39.62	10/11	122.8	79.7	125.5	78.2	5.9	67.2	338.5	−23.3	340.7	−25.4
13	JQ15A-F	14.02/39.90	4/9	199.4	−55.9	166.1	−59.2	6.8	185.2	159.4	−36.3	307.2	34.5
14	JQ16-17A-B	14.12/39.77	4/8	35.7	42.2	34.8	43.2	12.7	53.5	2.7	38.0	1.3	38.1
15	JQ18A-C	14.12/39.77	4/4	22.9	43.3	24.5	44.7	9.6	92.5	349.9	44.8	350.8	43.1
	Mean (all sites)		15	38.8, α_{95} = 7.8°	70.2, *K* = 24.9	44.4, α_{95} = 8.0°	72.6, *K* = 23.8			341.4, A_{95} = 11.6°	12.5, *K* = 11.9	341.7, A_{95} = 12.0°	7.4, *K* = 11.1
	Mean (sites: 1–12)		12	47.0, α_{95} = 6.3°	75.7, *K* = 48.8	61.2, α_{95} = 5.4°	76.5, *K* = 66.2			339.6, A_{95} = 11.2°	5.4, *K* = 16.0	342.1, A_{95} = 9.6°	−0.5, *K* = 21.6

Lat/Long, geographical latitude and longitude of the site; *N*/*n*, number of samples used in the mean/number of analyzed samples; Dec., declination; Inc., inclination; α_{95}, radius of the 95% confidence cone; *K*, precision parameter (Fisher, 1953); VPG, Virtual Geomagnetic Pole; Plong., Paleolongitude; Plat, Paleolatitude.

Table 2
Jequié paleomagnetic data for other components.

Site	Samples	Localization	*N*/*n*	Site mean directions						VGPs			
				Uncorrected		AMS corrected						AMS Corrected	
		Lat (°S)/Long (°W)		Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	α_{95} (°)	K	Plong. (°E)	Plat. (°N)	Plong. (°E)	Plat. (°N)
16	SM131-133	13.02/39.98	7/9	202.0	5.7	159.4	7.1	13.7	20.2	206.9	−66.0	73.1	−67.6
17	JQ13A-D	14.17/39.43	7/8	220.3	−9.0	135.6	−11.8	9.5	41.1	208.4	−45.9	72.1	−41.6
18	SM211-213	14.23/39.50	9/9	221.2	0.6	135.0	−2.1	8.3	39.7	215.2	−46.9	65.7	−42.9
19	SM221-223	14.27/39.40	7/7	225.7	4.3	131.6	4.1	13.3	21.4	219.9	−43.2	60.4	−40.7
20	JQ1-3	13.82/39.50	7/7	216.9	9.0	139.3	6.7	10.8	31.9	219.7	−52.5	61.3	−48.5
21	JQ4-6	13.83/39.50	6/7	15.4	−11.9	17.3	−9.2	11.0	37.9	24.7	73.0	23.8	70.6
22	SM171-173	13.98/39.93	5/8	26.3	−5.7	24.9	−10.8	11.0	49.4	29.3	61.7	33.3	64.0
	Mean		7	212.6	4.1			9.9	38.2	212.9 A_{95} = 9.0°	55.7 *K* = 46.3		
23	SM151-153	13.85/40.08	11/12	263.4	−41.2	93.5	−40.4	6.1	56.7	205.5	−0.1	73.1	2.2
24	SM181-184	14.13/39.75	15/15	274.6	−37.1	85.7	−38.3	6.2	39.2	211.0	9.1	70.3	9.1
25	SM201-203	14.20/39.65	11/12	253.8	−12.5	97.3	−12.2	3.3	191.2	220.0	−14.0	58.2	−5.5

Lat/Long, geographical latitude and longitude of the site; *N*/*n*, number of samples used in the mean/number of analyzed samples; Dec., declination; Inc., inclination; α_{95}, radius of the 95% confidence cone; *K*, precision parameter (Fisher, 1953); VPG, Virtual Geomagnetic Pole; Plong., Paleolongitude; Plat., Paleolatitude.

(eastern Brazil). These data together with new $^{40}Ar/^{39}Ar$ ages from Jequié rocks are used to evaluate the role of the Congo-São Francisco proto-craton in the formation of Atlantica and Columbia.

2. Geological setting and paleomagnetic sampling

The São Francisco craton (SFC), considered to be an extension of the West Congo craton (WC) in a pre-Atlantic configuration, is surrounded by Brasiliano-Panafrican belts formed between 630 and 550 Ma, and structurally marked by supracrustal units thrust over the cratonic domain (e.g., Teixeira et al., 2000; Alkmim, 2004). Exposures of the crystalline basement occur in the northern and southern parts of the SFC due to the extensive Neoproterozoic cover.

The tectonic scenario of the northern part of the SFC resulted from the agglutination of at least three major Archean blocks

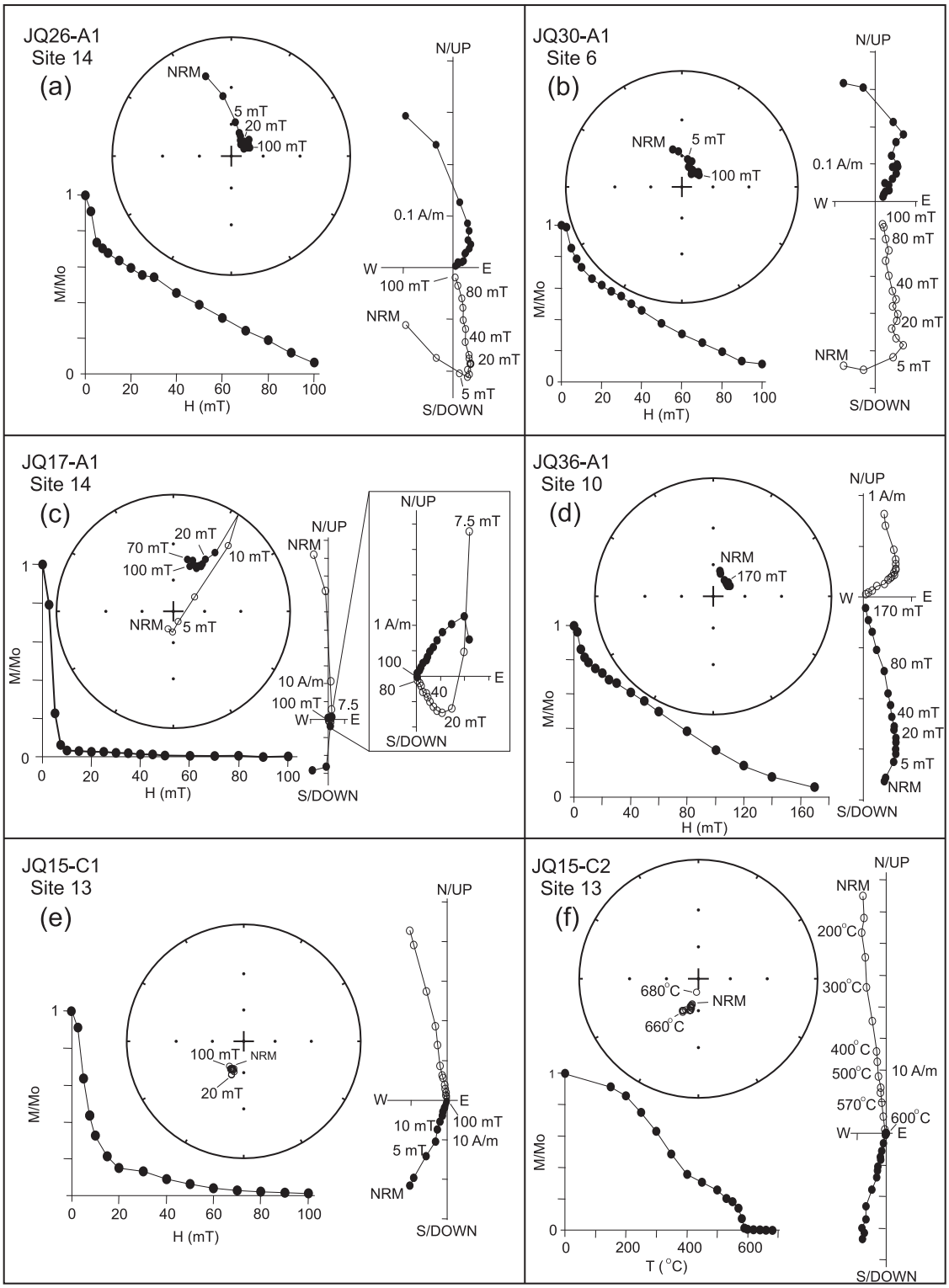


Fig. 6. Examples of AF and thermal demagnetization (component A). Stereographic projections (full (empty) symbols represent downward (upward) inclinations); orthogonal projections (full (empty) symbols represent horizontal (vertical) projections); magnetization intensity decay curves ($M/M_o \times H$).

(Serrinha, Jequié, Gavião; Fig. 1a), as well as coeval isolated granitoid nuclei (not shown in Fig. 1a), and intervening fragments (e.g., North Gabon, Chaillu massifs; WC). They assembled during Neoarchean/early Paleoproterozoic times (Barbosa and Sabaté, 2002, 2004) as a result of Paleoproterozoic orogeny – characterized by successive magmatic arcs from 2.3 to 1.9 Ga (e.g., Itabuna–Salvador–Curaçá collisional belt). Noteworthy each one of the Archaean blocks and related fragments exhibits distinct origins and tectonic histories, as demonstrated by the radiometric ages, isotopic signatures and geochemical characteristics of respective rocks. Such a Paleoproterozoic dynamics produced plutonic intrusions and led to widespread remobilization and metamorphism (amphibolite to granulite facies) over the Archaean domains at 2080–2050 Ma (Barbosa and Sabaté, 2002, 2004; Silva et al., 2002; Barbosa et al., 2008). Greenschist facies retrometamorphism related to this collisional belt is also reported, as indicated by K/Ar ages of ca. 1.9 Ga (Barbosa and Sabaté, 2003).

Much of the Jequié block (Fig. 1b) – the main focus of our work – is composed of mega-enclaves of supracrustal rocks and partly migmatised charnockite and enderbite gneisses which were originally named as Jequié Complex (Cordani and Iyer, 1978; Barbosa, 1990). Granitoid intrusions as old as 2.8–2.6 Ga are also present. The charnockite and enderbite correspond to calc-alkaline plutons (3.0–2.9 Ga), which have been intensely deformed and re-equilibrated in granulite facies during the Paleoproterozoic collisional event. The T_{DM} ages (3.4–3.0 Ga) indicate the heterogeneous protoliths from which the Jequié rocks derived, whereas the U–Pb ages (2800 Ma, 2700 Ma, 2640 Ma and 2500 Ma) reveal four major accretion/differentiation events (e.g., Marinho et al., 1994). Therefore the Jequié block was probably formed by a collage of primitive Archean arcs, coupled with reworked events, driven by oceanic crust subduction processes underneath the Gavião block. Along the NE sector of the Jequié block, charnockites and enderbites with protolith ages of 2473 ± 5 Ma (SHRIMP zircon) are also recorded whereas structural domes of charnockitic plutons in the northern sector of the Jequié block (see Fig. 1b) give U–Pb SHRIMP zircon ages of 2061 ± 6 and 2047 ± 14 Ma, and are contemporary with the collisional-related granulite-facies metamorphism (Silva et al., 2002). A similar age (2086 ± 18 Ma) was obtained for a granulite in the southern part of the block by the $^{207}Pb/^{206}Pb$ methodology (Ledru et al., 1994). After Early Proterozoic cratonization and uplift at around 2080–1900 Ma, the Jequié block became tectonically stable, as indicated by K–Ar amphibole and biotite ages up to 1900 Ma (Brito Neves et al., 1980). In addition, some K–Ar ages of ca. 1700 Ma may be influenced by the evolution of the Paramirim aulacogen (Fig. 1, inset) which occur along the NNW–SSE axis of the Craton (Cordani et al., 1992) westward from the Jequié block. K–Ar ages as young as 500–700 Ma were obtained in a region between the Jequié block and the adjacent Gavião block (Cordani et al., 1985). These ages probably resulted from thermal overprint and complete argon loss at ca. 600 Ma ago, related with the ultimate tectonic reactivation along the Paramirim province (see Fig. 1, inset).

The Itabuna–Salvador–Curaçá belt (Barbosa et al., 2003; Teixeira et al., 2000) is composed of low-K calc-alkaline tonalitic-trondhjemitic-granodioritic gneisses, heterogeneous granulites (including with shoshonitic affinity ones), and tholleitic mafic-ultramafic rocks, thought to represent ancient oceanic crust (Barbosa et al., 2008). They are covered by supracrustal rocks, such as mafic and intermediate metavolcanics, quartzites, banded iron formations and meta-ultramafics (Barbosa and Sabaté, 2002, 2003), which were later intruded by plutonic rocks (e.g., syenites, granites) between 2089 and 2098 Ma (Oliveira et al., 2004; Rosa et al., 2001).

In the present work, key-samples were collected along three transects in both the Jequié and Itabuna–Salvador–Curaçá domains (Fig. 1b) to investigate the paleomagnetic properties and the age of magnetization in high grade rocks by the use of $^{40}Ar–^{39}Ar$ method. Three to four block samples or three to five cores were extracted from each selected outcrop. Whenever possible, both solar and magnetic compasses were used for orienting samples; the difference between these two orientations was typically less than 6°.

Paleomagnetic and anisotropy of low-field magnetic susceptibility analyses (AMS) were performed in specimens from 62 oriented block samples and 22 cylindrical cores. Jequié charnockites and enderbites gneisses were sampled in sites 1–12 at the eastern sector, near Mutuípe town, and in sites 13 and 22 located along the road between Jequié and Ubaitaba (Fig. 1). In the same road, sites 14 and 15 are gneisses from the Ipiaú domain. The inner sector of the Jequié block, comprising predominantly granulites, is represented by site 23, near Jequié city, and site 16 sampled in a quarry 3 km from the highway BR116. The Itabuna–Salvador–Curaçá domain was sampled in sites along the Gandu-Ubaitaba road (sites 17, 20 and 21) and in some sites along the Jequié-Ubaitaba road (sites 18, 19, 24 and 25).

$^{40}Ar–^{39}Ar$ geochronology was performed in three rock samples. Two of them are from the Jequié block (near Mutuípe – site 5, and near Jequié – site 13) and the other (nearby Ubaitaba – site 18) is from the Itabuna–Salvador–Curaçá belt.

3. Paleomagnetism and magnetic anisotropy

3.1. Methods

Cylindrical cores were cut into 2.2 cm height specimens and submitted to conventional stepwise thermal and alternating field (AF) demagnetization to isolate the characteristic remanent magnetization (ChRM) component carried by the samples. Normally, steps of 2.5 mT (up to 15 mT) and 5 mT (15–100 mT) were employed for AF demagnetization using a tumbler Molspin AF demagnetizer, or an automated three-axis AF demagnetizer coupled to a 2G-Enterprises cryogenic magnetometer. Steps of 50 °C (from 100 °C up to 500 °C), and 20 °C (500–600 °C) were used for the thermal demagnetization in a Magnetic Measurements TD-48 oven. Measurements of remanent magnetization were carried out using a fluxgate spinner Molspin magnetometer or a 2G-Enterprises cryogenic magnetometer. Magnetic components were identified and calculated using orthogonal projections (Zijderveld, 1967) and principal components analysis (Kirshvink, 1980). At least 4 demagnetization steps were used to calculate vectors from least-squares

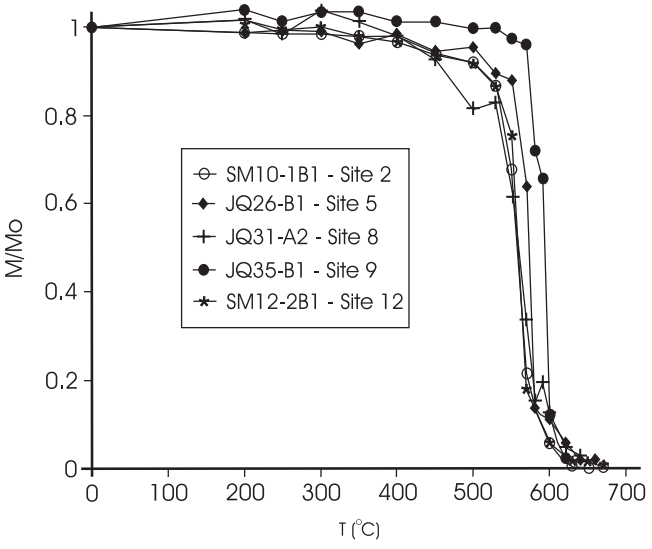


Fig. 7. Examples of normalized intensities (M/M_o) versus temperature for samples that yielded component A.

fits, and an upper limit for mean angular deviation (MAD) of 8° was set. Fisher's (1953) statistics was used to calculate site mean directions and the paleomagnetic pole. Magnetic mineralogy was investigated performing hysteresis curves with a Molspin vibrating sample magnetometer (VSM), by the acquisition of isothermal remanent magnetization (IRM) in a Magnetic Measurements pulse magnetizer, and using thermomagnetic curves preformed with a CS3 furnance coupled to a Kappabridge KLY-4S instrument (Agico, Brno). Anisotropy of low-field magnetic susceptibility (AMS) was measured to determine the magnetic fabric of the rocks using the same KLY-4S. Mean AMS eingenvectors (maximum susceptibility, K1, intermediate susceptibility, K2, and minimum susceptibility, K3) and 95% confidence regions were calculated using the bootstrap method of Constable and Tauxe (1990).

3.2. Magnetic mineralogy

Fig. 2 shows examples of thermomagnetic curves performed in air representing all sampled sectors. All samples, irrespective of rock type, show irreversible behavior and Curie temperatures around 580 °C, indicating Ti-poor titanomagnetite as the main magnetic carrier in the rocks. Also, the small Hopkinson peak on heating curves of some samples (e.g., JQ24D in Fig. 2) indicates SD-magnetite is present in these rocks. Most curves present a slight decrease during heating between 300 °C and 400 °C. This may indicate the presence of maghemite which is meta-stable and transforms to hematite during heating. Isothermal remanent magnetization (IRM) acquisition curves up to 1000 mT (Fig. 3) show only one step and completely saturates at fields of less than 400 mT,

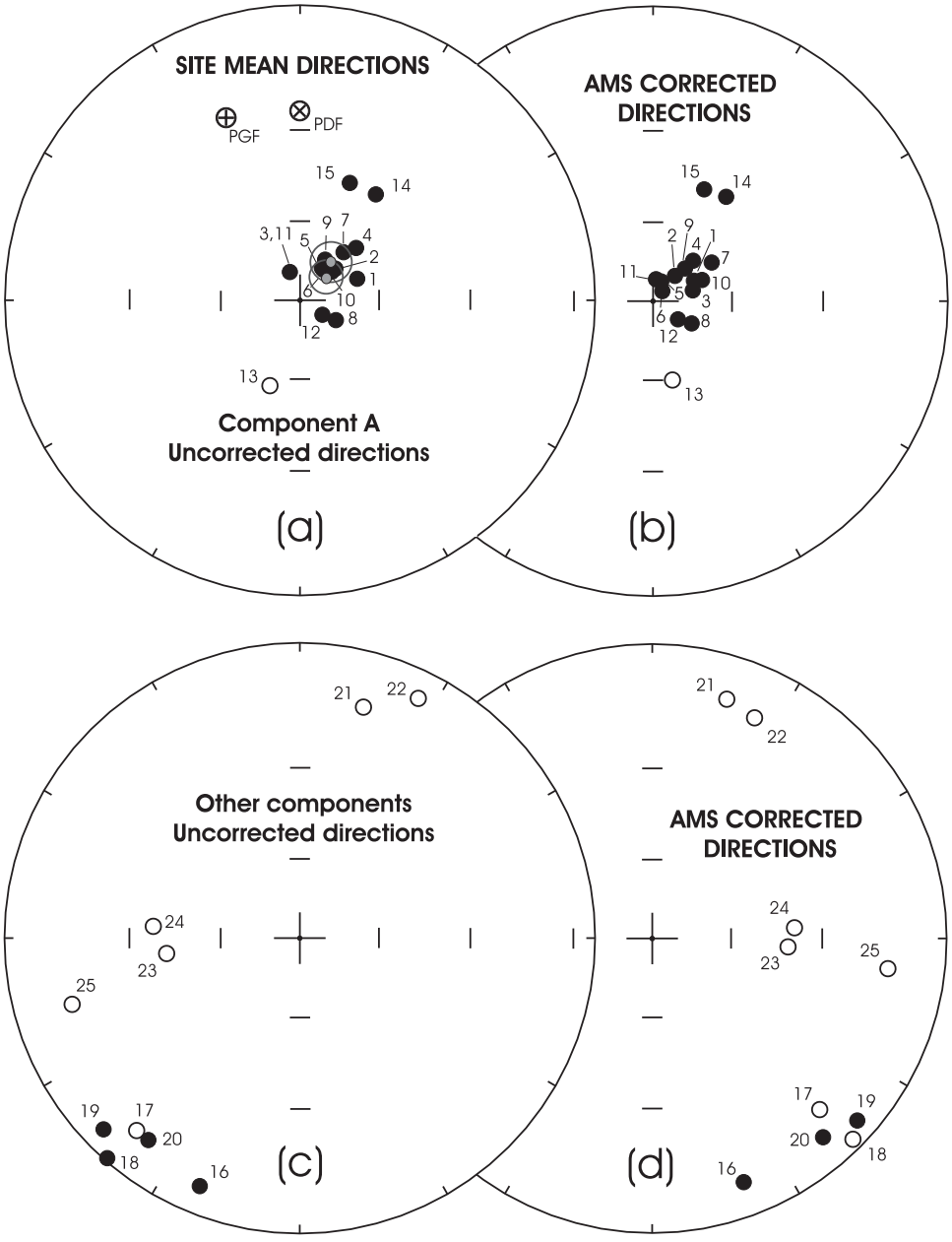


Fig. 8. Site mean directions for component A and for other components before (a and c) and after (b and d) AMS correction, respectively. Gray circles in (a) represent mean directions for component A; for all sites ($N = 15$) (the greatest α_{95}), and only for sites ($N = 12$) from the Mutuípe area. Full (empty) symbols represent downward (upward) inclinations. PDF, Present Dipolar Field; PGF, Present Geomagnetic Field. See text for AMS correction details. The numbers refer to the sites as listed in Tables 1 and 2.

which is typical of (titano)magnetite. Examples of hysteresis loops for samples from different domains are shown in Fig. 4, which depicts bulk coercive force (Hc), saturation magnetization (Js), and saturation of isothermal remanence (Jrs), after subtraction of the paramagnetic contribution from the high field portion of the curve. The coercivity of remanence (Hcr) was obtained after back-field measurements using the same Molspin VSM instrument. The coercivity of individual samples ranges from 1 to 18.5 mT, which is also typical of (titano)magnetite. Fig. 5 shows the Day's diagram (Day et al., 1977), which plots the Jrs/Js against the Hcr/Hc ratios for the studied samples. The limits between SD, PSD and MD fields and the SD–MD and SD–SP mixing lines are those proposed by Dunlop (2002a). All samples fall along a trend parallel to the SD–MD magnetite mixing curve with most samples plotting close to the MD field.

3.3. Paleomagnetism

Charnockites and enderbites gneisses from the eastern portion of the Jequié block presented a northeast downward (southwest upward) component after thermal and AF treatments (Table 1, Fig. 6). This direction, referred hereafter as component A, is characterized by high unblocking temperatures, between 540 °C and 600 °C, with hard-shoulder demagnetization patterns (Fig. 7). Directions obtained for the charnockites and enderbites near Mutuípe show steep inclinations. Outside Mutuípe, directions obtained on gneisses of the Ipiaú domain show slightly shallower inclinations for two sites (sites 14 and 15, Fig. 6c). Charnockite of Site 13 shows an opposite polarity, with directions pointing steep upward to the southwest (Fig. 6e and f). The site mean directions for component A (sites 1–15) are plotted in Fig. 8a, and two

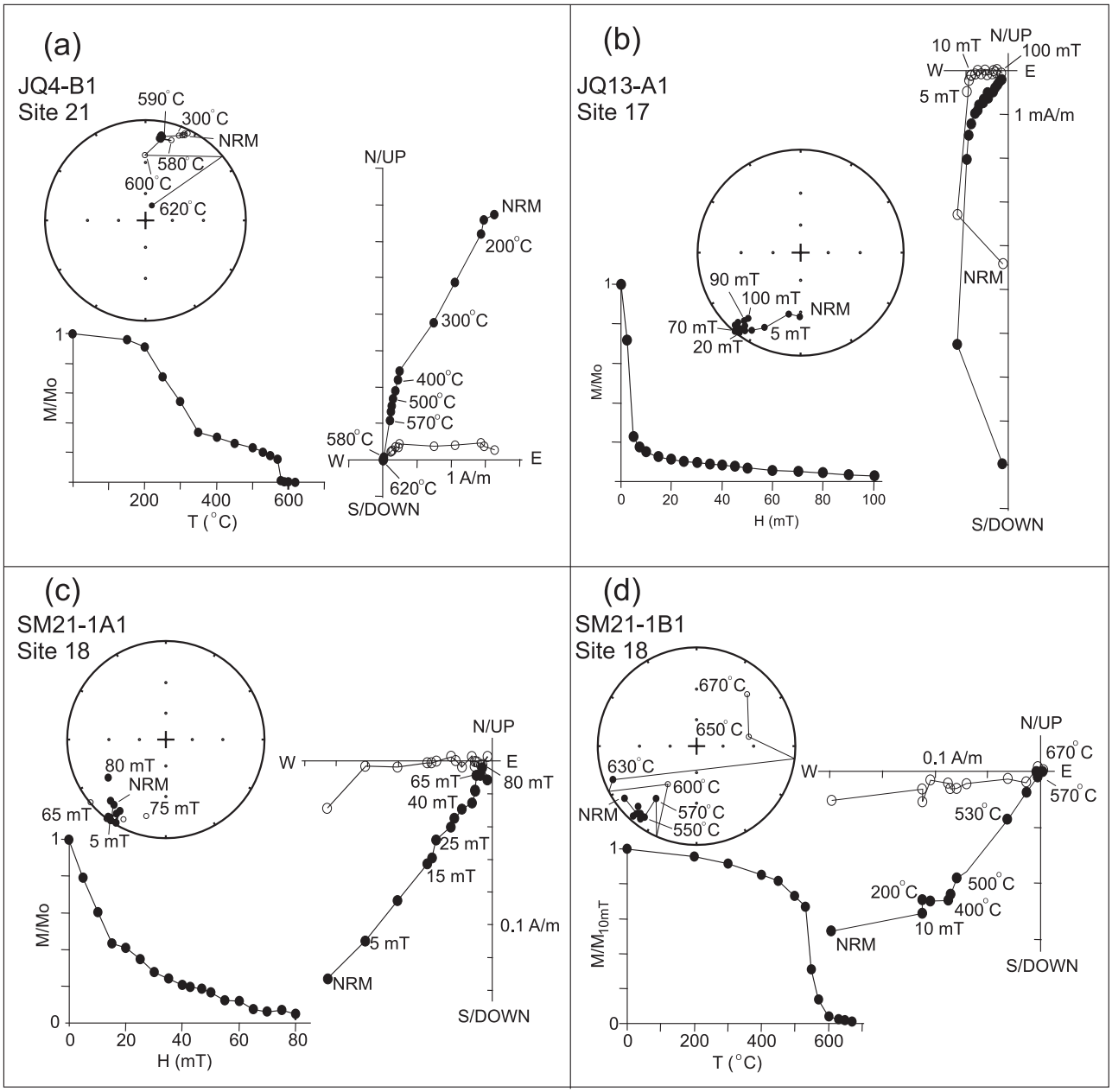


Fig. 9. Examples of AF and thermal demagnetization for samples with southwestern or northeastern, shallow inclinations. Stereographic projections (full (empty) symbols represent downward (upward) inclinations); orthogonal projections (full (empty) symbols represent horizontal (vertical) projections); magnetization intensity decay curves ($M/M_o \times H$).

mean directions were calculated: one direction including all sites, and all rock-types that carry the component A ($N = 15$), $D_m = 38.8^\circ$, $I_m = 70.2^\circ$ ($\alpha_{95} = 7.8^\circ$, $K = 24.9$), and a second one using only the sites from the charnockites and enderbites from the Mutuípe region ($N = 12$); $D_m = 47.0^\circ$, $I_m = 75.7^\circ$ ($\alpha_{95} = 6.3^\circ$, $K = 48.8$). These mean directions are similar at confidence levels (gray full circles in Fig. 8a) but the mean that includes sites outside Mutuípe shows slightly higher dispersion of directions. They yielded paleomagnetic poles located at 341.4°E, 12.5°N ($A_{95} = 11.6^\circ$; $K = 11.9$) and 339.6°E, 5.4°N ($A_{95} = 11.2^\circ$; $K = 16.0$), respectively.

Sites from the inner Jequié block or located in the Itabuna–Salvador–Curaçá belt (see Fig. 1), comprising migmatites, heterogeneous granulites, tonalities, trondhjemites and gneisses, yielded either a northeastern/southwestern, low inclination component (sites 16–22), or a western, shallow upward component (sites 23–25). Examples of AF and thermal demagnetization of the first most representative component are shown in Fig. 9. Site mean directions are presented in Fig. 8c and Table 2. The reversal test of McFadden and Lowes (1981) indicates that normal and reversed directions do not share a common mean at the 95% confidence level, but a positive reversal test with classification C was disclosed by the McFadden and McElhinny (1990) test. So, a mean direction was calculated using reverse and normal site mean directions ($D_m = 212.6$; $I_m = 4.1$; $\alpha_{95} = 9.9^\circ$; $K = 38.2$), which yielded a paleomagnetic pole located at 55.7°S; 212.9°E ($A_{95} = 9.0$; $K = 46.3$).

3.4. Anisotropy of low-field magnetic susceptibility (AMS)

Jelinek, 1981; Samson and Alexander, 1987; Table 1 (supplementary data file) presents AMS data for the analyzed sites. The mean magnetic susceptibility, expressed by $K_m = (K_1 + K_2 + K_3)/3$ (in SI units) shows a strong variability, ranging from 10^{-4} to 10^{-1} SI with a clear bimodal distribution (Fig. 10a). The first group is characterized by lower susceptibilities with a peak value around 2.5×10^{-3}. The second group is characterized by higher susceptibilities and a peak value around 3×10^{-2}. Samples from charnockites and enderbites, which show the component A, fall into both groups, therefore this behavior is not directly related to the rock types but to local heterogeneities among the sampled rocks. The degree of anisotropy ($P = K_1/K_3$) for all rock types is often very high with site-mean P values reaching up to 1.881, with a peak of distribution around 1.3–1.4 (Fig. 10b). The P versus K_m plots shows a clear increase of the degree of anisotropy (P) with the increase of the mean susceptibility (K_m) (Fig. 11). The shape of anisotropy ellipsoids, estimated from the T parameter ($T = [2\ln(K_{int}/K_{min})/\ln(K_{max}/K_{min})]^{-1}$), is dominantly oblate ($T > 0$) for all rock types and magnetic components (Fig. 11).

Fig. 12 shows the distribution of AMS principal axes for the studied samples. They were separated according to the magnetic components and geographic distribution of sites. We note that the structural trend at the border of the Jequié block changes from NW–SE in the Mutuípe sector, to NE–SW in the South. Sites with component A are shown in Fig. 12a, while sites with other components are shown in Fig. 12b. Sites located around the town of Mutuípe (1–12) are shown in Fig. 12c, whereas sites outside the Mutuípe area (16–25) are shown in Fig. 12d. From these stereoplots it is clear that the orientation of the principal axes of anisotropy is chiefly controlled by the trend of the metamorphic fabric (Barbosa and Sabaté, 2002, 2004). Magnetic foliations in the northern sector of the Jequié block show NW-trending directions and moderate inclinations that are in accordance with the regional metamorphic foliation trend. This fabric pattern is also reproduced in Fig. 12a since most of the sites with component A are found in samples from charnockites and enderbites gneisses from the Mutuípe region. Magnetic foliations in the south are dominantly

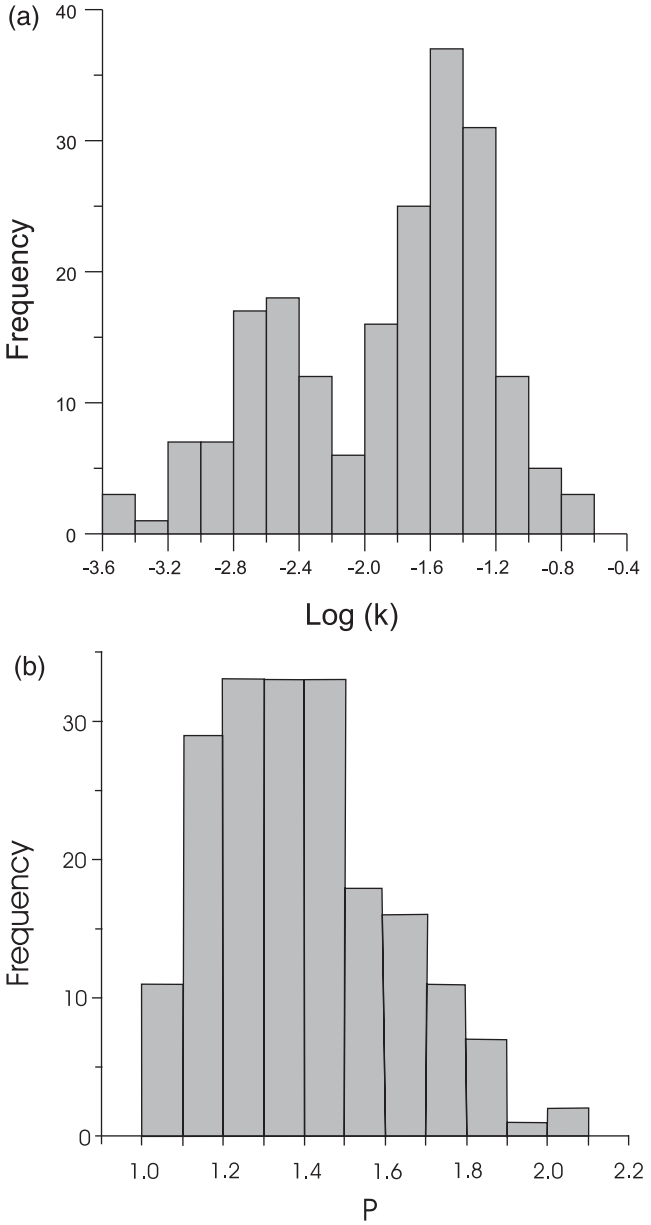


Fig. 10. (a) Frequency of the logarithm of mean specimen susceptibility ($K_m = [K_1 + K_2 + K_3]/3$) showing a bimodal behavior. (b) Frequency of the degree of anisotropy ($P = K_1/K_3$).

vertical, NE-trending and are consistent with the structure of the Paleoproterozoic Itabuna–Salvador–Curaçá belt. Only three sites in the southern region show NW-trending foliation (gray symbols in Fig. 12d): sites 14 and 15 that show the component A, and site 24. Note that these sites are close to the contact of the Itabuna–Salvador–Curaçá belt with the Ipiaú domain of the Jequié block (Fig. 1). In both regions, magnetic lineations plunge shallowly NNE or NNW. The strong correlation between the magnetic fabric and the structural grain of both the Jequié block and the Itabuna–Salvador–Curaçá belt suggests that the magnetic fabric in the analyzed rocks was imposed during the ~2.0–2.1 Ga collision leading to the final architecture of Jequié and other Archean blocks.

3.5. AMS correction of paleomagnetic directions

We note that the strong magnetic anisotropy observed in the studied rocks was imposed by sample gneissosity acquired during deformation and metamorphism of the Jequié block and in the

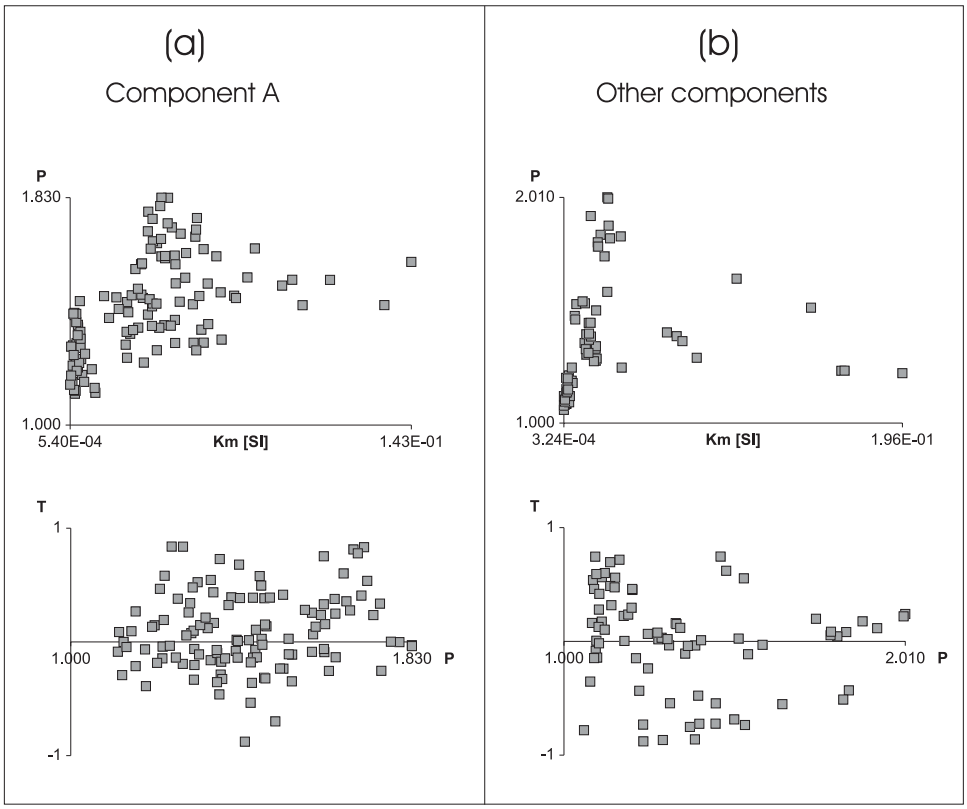


Fig. 11. P versus K_m plot and Jelinek's parameter ($T = [2\ln(K_{int}/K_{min})/\ln(K_{max}/K_{min})]^{-1}$) versus P plot for all samples according to magnetic component: (a) A component and (b) other components.

Itabuna–Salvador–Curaçá Belt. Consequently, the paleomagnetic directions acquired during or after fabric formation would reflect transformation induced by the magnetic anisotropy (Raposo et al., 2003). In order to correct the magnetic directions for the anisotropy effects we used the equation $[H] = [K^{-1}]\,[M]$, where K is the mean anisotropy tensor of a site, M is the mean remanence vector isolated in that site, and H is the mean direction for that site after correction (representing the orientation of the paleofield).

Since we intended to correct a thermoremanent component acquired during cooling of the Jequié block, ideally we must have performed measurements of anisotropy of thermoremanence (ATRM) (Jackson, 1991). But this is not possible since the magnetic mineralogy of the samples is strongly altered during heating, as attested by the non reversible behavior of thermomagnetic curves (Fig. 2). Alternatively, we can use other magnetic anisotropies to perform the correction. We have chosen the non-destructive and time-effective anisotropy of low-field magnetic susceptibility (AMS). In doing so we assume that the magnetic susceptibility is almost completely controlled by the remanence carriers (magnetite) and the magnetic anisotropy ellipsoid is close to that of the thermal remanent anisotropy in orientation and shape. The first assumption is substantiated by the fact that the magnetic susceptibility of samples is very high ($>10^{-2}$ SI) and consequently controlled by the magnetite content. The second assumption is less well constrained. The orientation of AMS and remanence ellipsoids are usually close except when prolate SD grains are present in significant amounts (e.g., Rochette et al., 1992). Given the coherence of magnetic fabrics throughout the different sectors of the Jequié block, we consider that the orientation of the AMS ellipsoid is not affected by inverse fabric phenomena. In magnetite-bearing rocks, the anisotropy degree of the remanence is usually higher than that of the magnetic susceptibility (Jackson, 1991). Thus, the correction based on AMS must be considered as a conservative estimate of the anisotropy effect. For instance, a detailed paleomagnetic work on high-grade rocks with high anisotropy degree was performed in southeastern Brazil by Raposo et al. (2003). In this work, the correction based on AMS was compared to the corrections based on the anisotropies of anhysteretic remanence (AAR) and isothermal remanence (AIRM). The results show small differences in the final corrected paleomagnetic pole thus attesting the validity of AMS for correcting poles derived from high-grade metamorphic rocks.

The corrected site mean directions (and corresponding virtual geomagnetic poles) are presented in Tables 1 and 2, and plotted in Fig. 8b and d. New corrected mean direction and pole were calculated for all sites ($N = 15$) with component A. Although the dispersion of the mean direction calculated for all sites is still low and could easily reflect the secular variation of the geomagnetic field we preferred to exclude sites 13–15 from the mean, since the corresponding pole is significantly improved when compared with the uncorrected one, as shown by the statistic parameters α_{95} and K (Table 1). The mean direction calculated for the Mutuípe area is $D_m = 61.2^\circ$, $I_m = 76.5^\circ$ ($N = 12$, $\alpha_{95} = 5.4^\circ$, $K = 66.2$) which yielded a pole at 342.1°E, −0.5°N ($A_{95} = 9.6^\circ$).

The AMS correction for the other sites resulted normally in significant changes in direction (Fig. 8d). The southwestern, low inclination component changed to southeastern, low inclination directions after AMS correction. The sites with a shallow to moderate negative inclination component changed from western to eastern directions after AMS correction. The fact that after AMS correction directions disperse suggests that correction is probably no applicable here. The more complicated results from the southern Jequié block may be a reflex of the younger Paramirim Poliphase event as described above in the geological setting. This is corroborated by our ^{40}Ar–^{39}Ar dating of site 17 from the Itabuna/Curaçá/Salvador belt whose perturbed spectra suggest that some younger tectonic event occurred in the region (see below).

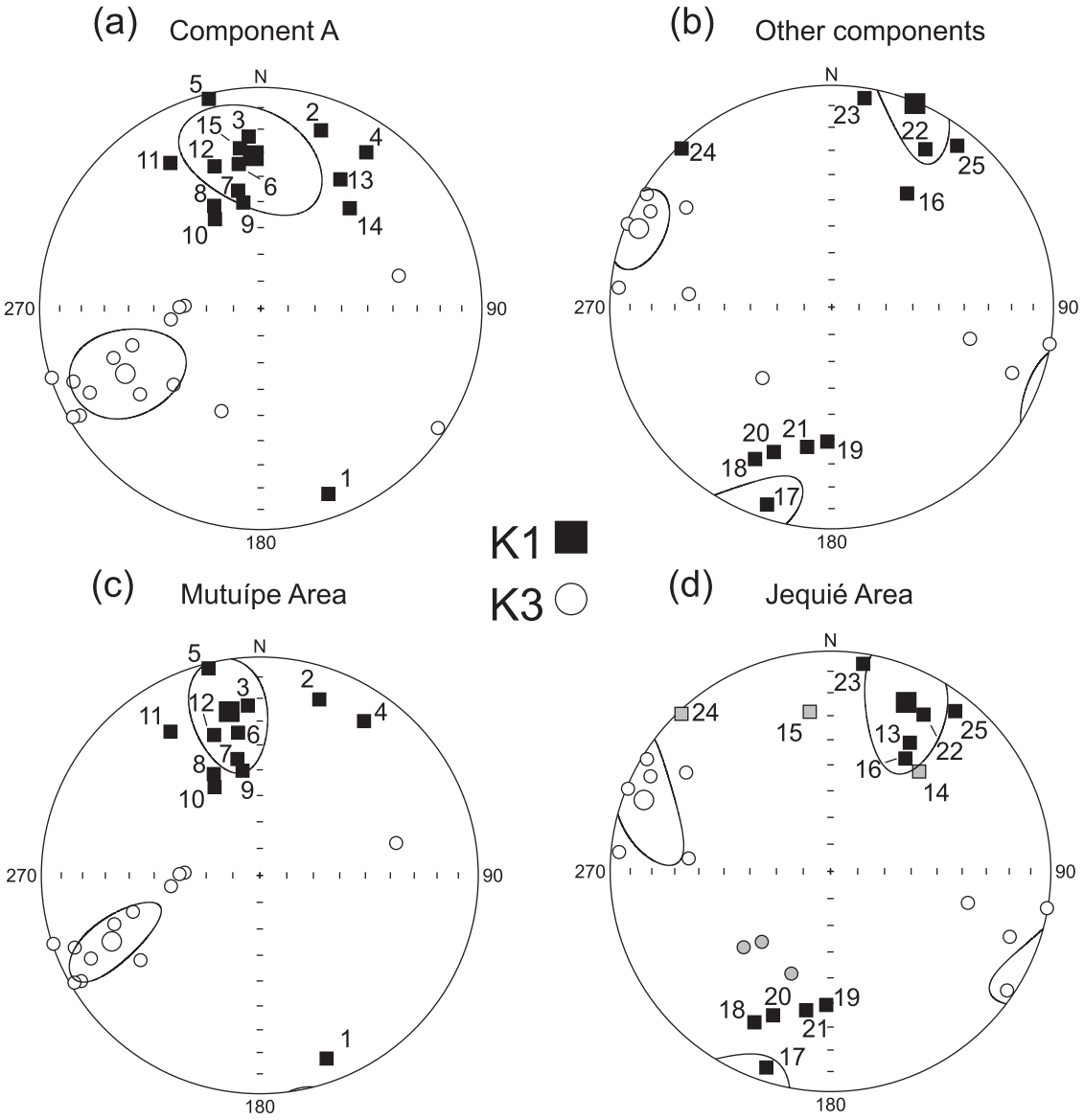


Fig. 12. Site mean anisotropy of low-field magnetic susceptibility (AMS) according to magnetic component: (a) A component, (b) other components, and according to the sampled area: (c) Mutuípe area, and (d) outside Mutuípe area (light gray symbols indicate sites (14, 15 and 24) with NW-trending magnetic foliations). Only the axes of maximum anisotropy (K_1) and minimum anisotropy (K_3) are shown. The numbers refer to the sites as listed in Tables 1 and 2.

4. ^{40}Ar–^{39}Ar geochronology

We have carried out ^{40}Ar–^{39}Ar dating to address the cooling history of the investigated rocks from the Jequié block and the adjacent Itabuna–Salvador–Curaçá belt. Two samples of charnockite gneisses from the eastern part of the Jequié block that present the component A were selected for dating. Sample JQ25-A (site 5) from the Mutuípe area presents a steep, positive inclination magnetic component. Sample JQ15-D1 (site 13), collected ca. 60 km southward, is the only site where an inverse polarity component A (negative inclination) was observed. The third sample (site 18; sample JQ13-D3), collected near Ubaitaba, displays a distinct, peculiar magnetic component. Hornblende and biotite grains from the first two samples were dated, whereas only hornblende grains from sample JQ13-D3 were analyzed. Methodology and results are shown in Table 2 (supplementary data file).

^{40}Ar–^{39}Ar analysis performed on amphibole grains from sample JQ25-A (site 5) yielded a plateau age of 2035 ± 4 Ma for 88.1% of released Ar (Fig. 13b). Biotite grains from the same sample yielded a plateau age of 1876 ± 5 Ma, defined by 55.2% of released Ar (Fig. 13d). Amphiboles from sample JQ15-D1 (site 13) did not define plateau ages, and the two replicate analyses yield integrated (total gas) ages of 2023 ± 3 Ma and 2016 ± 3 Ma (Fig. 14a and b). Biotites from the same sample yielded a plateau age of 1766 ± 3 Ma, defined by 81.2% of released Ar gas (Fig. 14d). ^{40}Ar–^{39}Ar hornblende analysis performed on sample JQ13-D3 (site 17) produced perturbed spectra (Fig. 15a and b) and did not yield plateau ages. The integrated ages 1468 ± 9 Ma and 1574 ± 15 Ma, respectively, are probably meaningless, if compared with the geochronological pattern recorded by the Paleoproterozoic belt.

5. Discussion

5.1. Age of the Jequié pole

The age and origin of magnetization in polymetamorphic rocks is complicated by the uncertainty over the mechanism of magnetization, i.e. either thermoremanent (in pre-existing grains) or chemical, carried by a newly formed ferromagnetic phase. Since magnetite appears to be the chief magnetic carrier, the origin

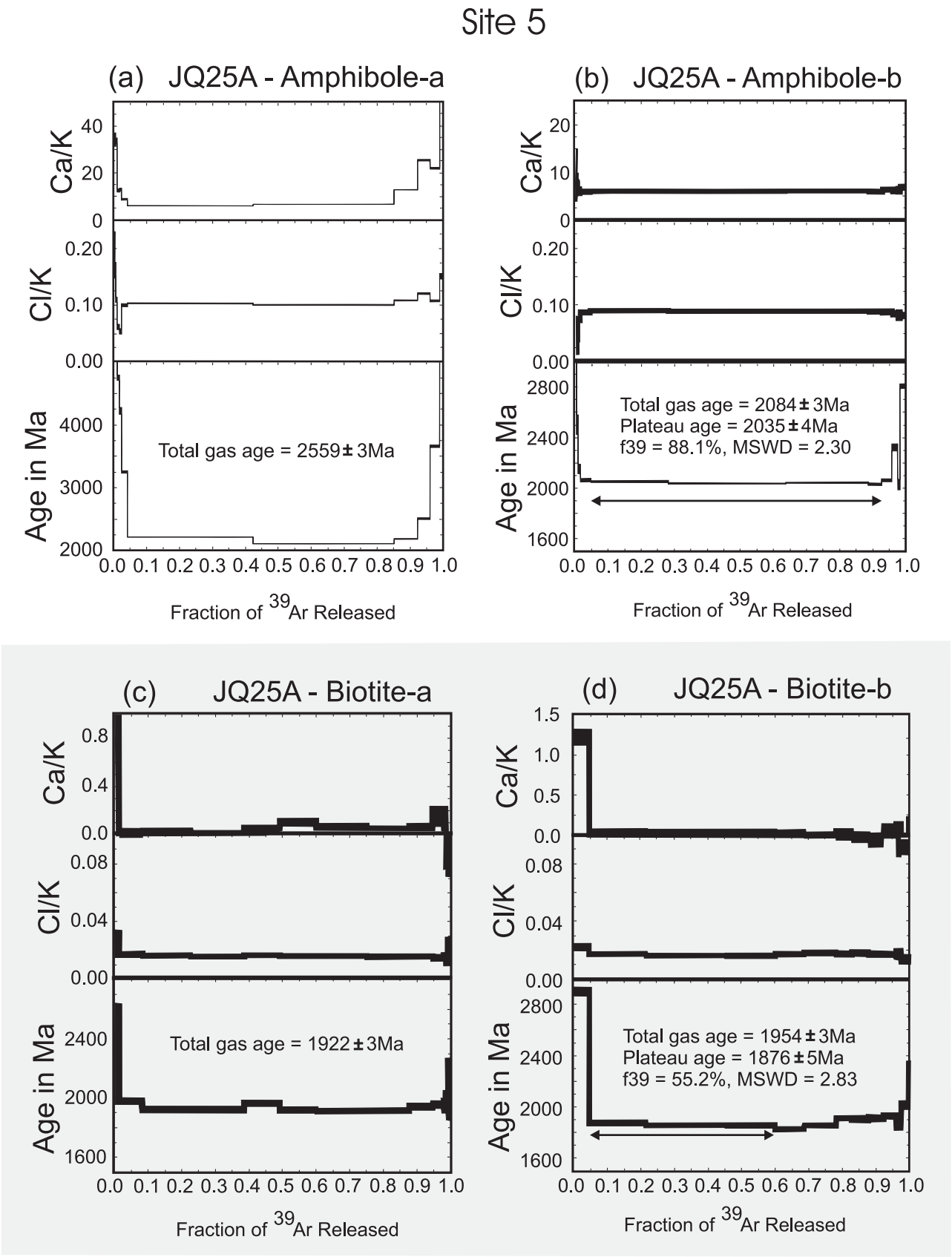


Fig. 13. Argon degassing spectra for sample JQ25A – (a and b) hornblende, (c and d) biotite. Arrows on age spectra indicate the individual steps used to calculate plateau ages. Plateau ages reflect five or more consecutive steps representing >50% of the total gas whose ages overlaps within errors at the 2σ level. Ar–Ar methodology and data are detailed in Table 2 (supplementary data file).

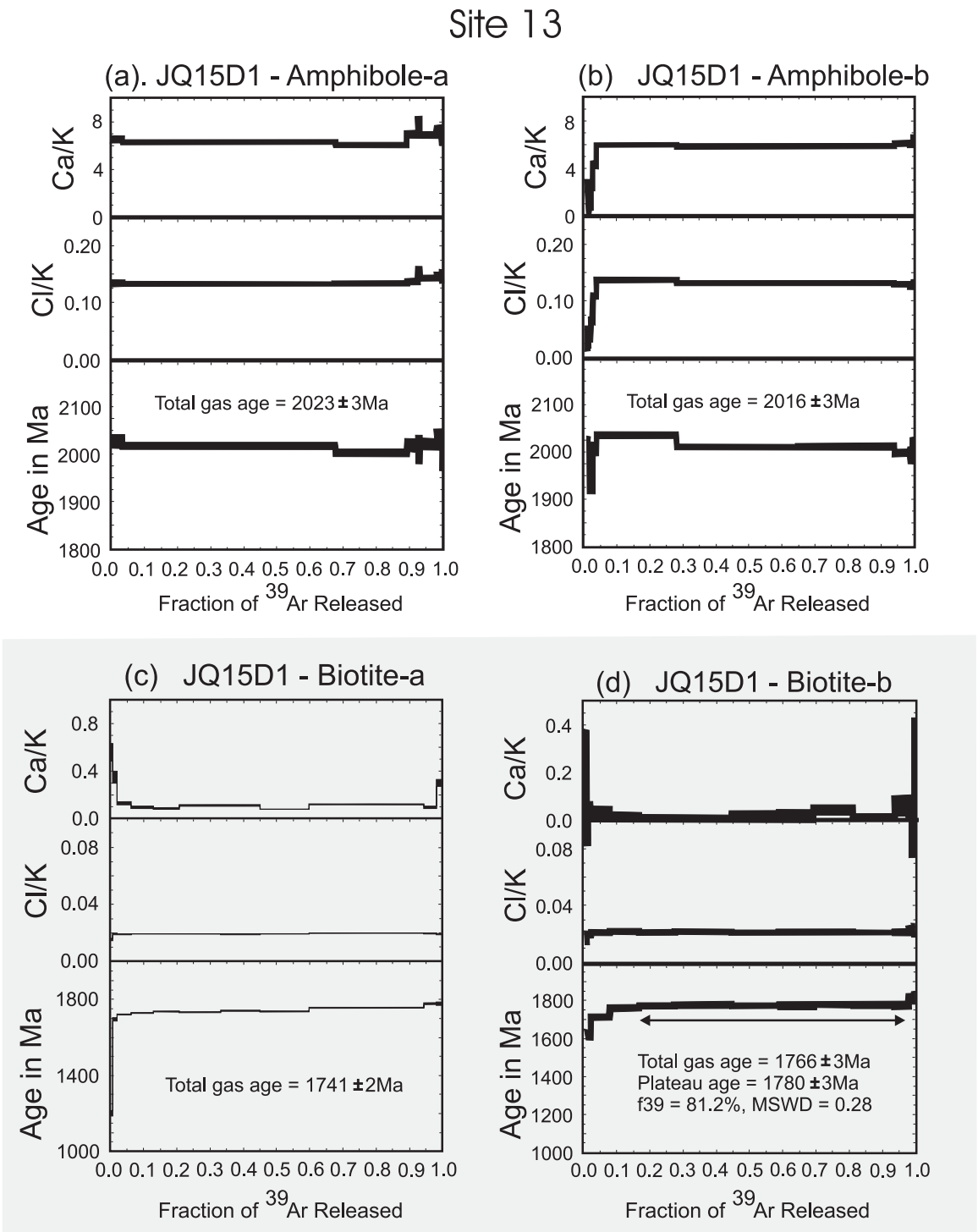


Fig. 14. Argon degassing spectra for sample JQ15D1 – (a and b) hornblende, (c and d) biotite. Arrow on age spectra indicates the individual steps used to calculate plateau age. Plateau ages reflect five or more consecutive steps representing >50% of the total gas whose ages overlaps within errors at the 2σ level. Ar–Ar methodology and data are detailed in Table 2 (supplementary data file).

of this oxide is critical in identifying the characteristic remanent magnetization as thermal or chemical in origin. In quenched lavas, Ti-rich magnetite is common, but slower cooling allows the ülvospinel-magnetite component to unmix by oxyexsolution at temperatures below 500 °C (Ghiorso, 1997), with the ulvospinel component oxidizing to form ilmenite at normal oxygen fugacities (Frost, 1991). At the elevated temperatures of granulite metamorphism, enhanced diffusion allowed magnetite and ilmenite to form discrete, unmixed phases. The ubiquity of magnetite in the diverse granulitic rock types studied here suggests that magnetite was the end-product of high temperature reactions, with magnetization acquired during monotonic cooling. Previous radiometric data on Jequié charnockites indicate that the peak granulite metamorphic facies was imposed at around 2.0–2.1 Ga ago (U–Pb SHRIMP ages of 2061 ± 6 Ma and 2047 ± 14 Ma). Our radiometric ages therefore mark the cooling of the region during tectonic exhumation.

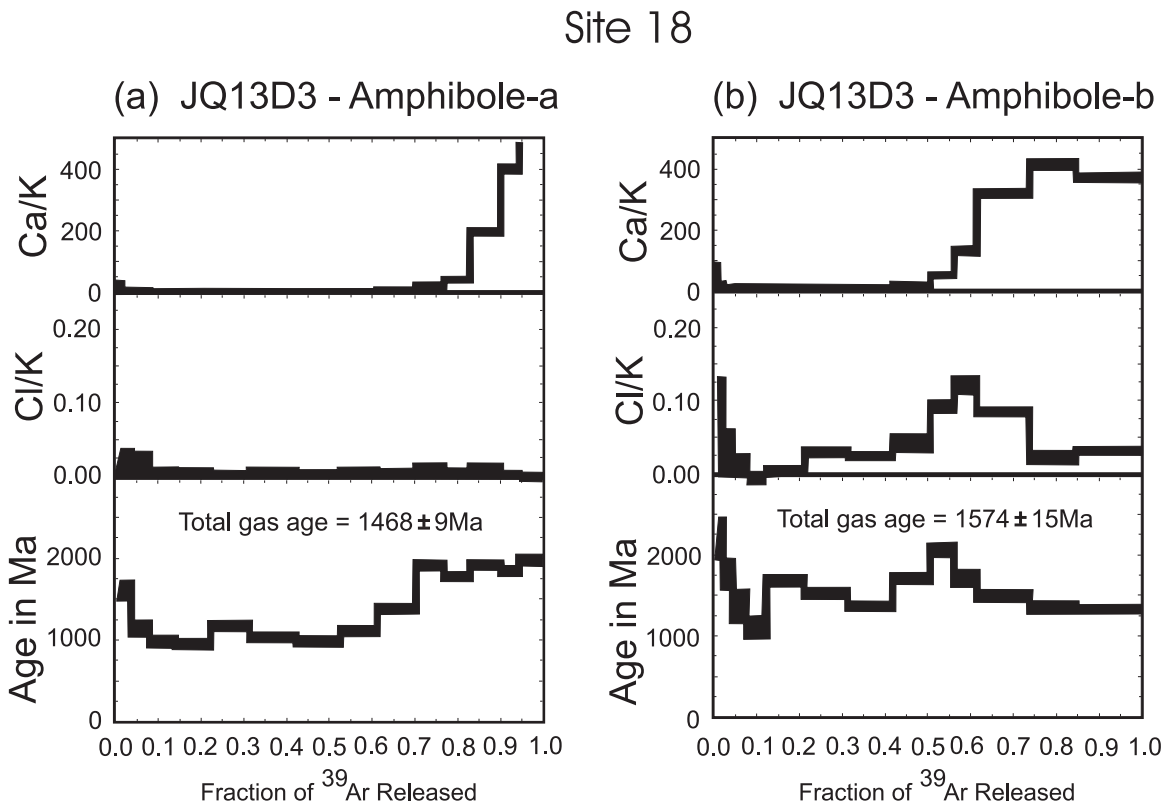


Fig. 15. Argon degassing spectra for sample JQ13D3 – (a and b) hornblende. Ar–Ar methodology and data are detailed in Table 2 (supplementary data file).

Theoretical consideration provide a technique for equating fast laboratory unblocking temperatures of thermal remanences with slow, cooling of metamorphic mobile belts by using ^{40}Ar–^{39}Ar ages. These corrected unblocking temperatures can then be compared with the isotopic closure temperatures of the geochronological markers. In Mutuípe, the ^{40}Ar–^{39}Ar cooling ages obtained on amphibole (closing temperature of 550 ± 30 °C, Dahl, 1996) and biotite (blocking temperature of 325 ± 25 °C, McDougall and Harrison, 1999) from sample JQ25-A (site 5 – Mutuípe area) imply a very low cooling rate of ~1.4 °C/Ma between 2035 ± 4 Ma and 1876 ± 5 Ma. The unblocking temperatures in laboratory (1 h heating) for our samples are between 540 °C and 600 °C. These temperatures correspond to unblocking temperatures between 502 °C and 600 °C for a heating time in nature in the order of 10^7 years, according to Pullaiah et al.'s (1975) equation. Taking these corrected unblocking temperatures and considering the cooling rate of 1.4 °C/Ma between amphibole and biotite ages, the age of magnetization would be comprised between 1985 and 2089 Ma (Fig. 16). The increasing tilt of the tie-lines at lower laboratory blocking temperatures in the nomogram by Pullaiah et al. (1975) suggest that the age of magnetization is closer to the youngest estimate.

5.2. Paleoproterozoic record of Congo-São Francisco Craton and neighboring plates

Paleoproterozoic paleomagnetic poles from Congo-São Francisco Craton are few in number (Table 3). Although some VGPs obtained for granulites from the Salvador area – SFC (D'Agrella-Filho et al., 2004) and for basement rocks from Gabon – Congo craton (D'Agrella-Filho et al., 1996) – are attributed to the Paleoproterozoic, the ages of these poles are still poorly constrained. Better paleomagnetic results were obtained for the Uauá dike swarm (D'Agrella-Filho and Pacca, 1998), although its estimated age is similarly poorly constrained by Rb/Sr mineral isochrons (1.90–1.98 Ga; Bastos Leal et al., 1994). The Jequié pole, determined in here after anisotropy correction is the best result available for the Paleoproterozoic in the Congo-São Francisco Craton. One must be concerned about an eventual tilting of the region after remanence acquisition by the high-grade metamorphic rocks of the Jequié block. However, the fact that most of the regional deformation has occurred during the Paleoproterozoic, with very limited deformation occurring in the Meso- to Neoproterozoic, and the similarity in metamor-

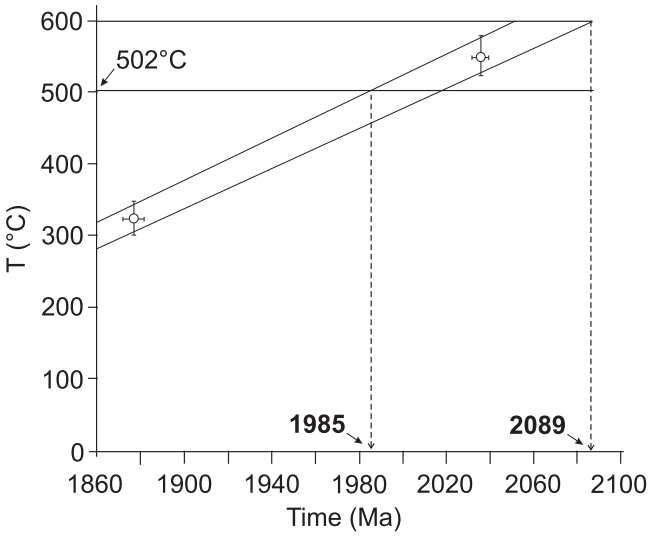


Fig. 16. Cooling history of the Jequié block obtained from $^{40}Ar/^{39}Ar$ isotopic ages for sample JQ25A. The calculated range in geologically magnetic blocking temperatures (502–600 °C) is also shown along with the intersection of those within the cooling curve. Collectively, they provide an estimate of the upper and lower bounds of the magnetic age. Error bars reflect the uncertainty in age and closure temperature of the isotopic systems as described in the text.

Table 3
Selected paleomagnetic poles for Congo-São Francisco, Kaapvaal and Amazonian cratons.

Formation	Code	Age (Ma)	Plat (°N)	Plong (°E)	D_p/A_{95} (°)	D_m (°)	Q-factor	Ref.
Kaapvaal								
Post Waterberg Dolerite	WSD	1879–1872	15.6	17.1	/8.9		5	1
Phapaborwa Complex 2	PB2	1880–1900	4.2	357.8	8.8	8.8	3	2
Mashonaland sills	MASH	1830 ± 230	7.6	338.2	5.1	5.1	5	3, 4
Post-Bushveld dolerite dykes	PBD	~1900 or 1649 ± 10	8.7	22.0	18.0	20.6	4	5
Mamatwan-type ore 2	MAM-2		12.1	321.8	3.4	6.0	2	6, 7
Sand River dykes	SRD	1876 ± 68	2.3	9.1	10.3	10.3	4	8
Limpopo metamorphics "A"	LMA	1950–1980?	26.1	22.3	7.9	10.3	3	8, 9
Witwatersrand overprint	WITS	1945 ± 40	19.1	45.6	7.8	7.8	2	10
Upper Swaershoek/Alma Fs.	WUBS-II	1930–2054	−10.5	330.4	9.8	9.8	6	6
Vredfort mean	VRED	2023 ± 4	22.8	41.6	10.6	10.6	5	11
Hartley Lavas	HAR	1928 ± 4	12.5	332.8	16	16	4	12
Lower Swaershoek F.	WUBS-I	2054 ± 4	36.5	51.3	10.9	10.9	5	6
Phapaborwa Complex 1	PB1	2059–2061	35.9	44.8	6.9	10.5	3	2, 13, 14
Bushveld middle complex	BMC	2060 ± 2	20.0	33.0	5.0	5.0	5	15
Bushveld Upper complex	BUC	2061 ± 27	16.0	32.0	11.0	11.0	4	15,16
Bushveld main and upper zones	BMUZ	2061 ± 27	11.5	27.1	4	4	6	12
Congo								
Gabon metamorphics	GM	1900–2200	19	44	29.8	29.8	2	17
São Francisco								
Uauá dikes	UD	1900–1980	23.8	331.4	/6.5		3	18
Jequié Complex	JQc	1985–2089	−0.5	342.1	/9.6		4	19
Salvador granulites – normal	SMN		28.2	209.0	/11.7		3	20
Salvador granulites – reverse	SMR		30.4	344.7	/26.7		2	20
Amazonia								
Mg–K granite	TUMU	2050–2070	18.9	273.7	19.2	22.3		21
Granite	TAMP03	2050–2070	−6.9	300.1	15.9	16.1		21
Granite	MATA02	2050–2070	14.9	289.2	40.6	42.7		21
Granite	APPR02	2050–2070	4.5	298.9	19.1	19.2		21
Granodiorite	APPR05	2050–2070	−5.9	296.9	34.3	35.1		21
Granite	APPR06	2050–2070	−18.5	294.3	21.3	23.0		21
Granite	APPR08	2050–2070	5.3	293.4	16.8	17.2		21
Mean	GF_1	2050–2070	1.8	292.5	/11.9			19
Armontabo River Tonalite	ARMO	~2030	−2.7	346.3	14.2	16.9	4	21
Oyapok granitoids and volc.-sed.	OYA	2016–2024	−28.0	346.0	/13.8		4	21,22
Imataca granulite	IM_1	1960–2080	−49.0	18.0	/18.0		1	23
Imataca granulite	IM_2	1960–2080	−29.0	21.0	/18.0		1	23
Encrucijada granites	EnA1	1968–1976	−55.0	8.0	/6.0		1	24
Encrucijada granites	EnA2		−37.0	36.0	/18.0		1	24
North Guiana granites	PESA	~1970	−56.7	25.1	6.2	12.4		21
North Guiana granites	ROCO	~1970	−58.0	26.4	7.9	15.8		21
North Guiana granites/granodior.	MATI	~1970	−58.6	25.5	9.7	19.4		21
North Guiana granites	ORGA	~1970	−59.7	44.7	10.1	19.5		21
Mean	GF_2	~1970	−58.5	30.2	/5.8		4	19
Rio de la Plata Craton								
Soca and Isla Mala granites	SMG	2050–2070	−14.6	279.7	8.2	9.1	4	25

Plat, Paleolatitude; Plong, Paleolongitude; *Dp* and D_m, semiaxes of the cone of 95% confidence about the pole; A_{95}, semiangle of the cone of 95% confidence about the pole; Q-factor, reliability of the pole according to Van der Voo (1990).
References: 1, Hanson et al. (2004); 2, Morgan and Briden (1981); 3, McElhinny and Opdyke (1964); 4, Bates and Jones (1996); 5, Letts et al. (2005); 6, de Kock et al. (2006); 7, Evans et al. (2001); 8, Morgan (1985); 9, Buick et al. (2006); 10, Layer et al. (1988); 11, Salminen et al. (2009); 12, Evans et al. (2002); 13, Reischmann (1995); 14, French et al. (2002); 15, Hattingh (1989); 16, Hattingh (1999); 17, D'Agrella-Filho et al. (1996); 18, D'Agrella-Filho and Pacca, 1998; 19, This work; 20, D'Agrella-Filho et al. (2004); 21, Théveniaut et al. (2006); 22, Nomade et al. (2001); 23, Onstott and Hargraves (1981); 24, Onstott et al. (1984); 25, Badgen et al. (2009).

phic retrograde evolution across the studied region (Barbosa and Sabaté, 2002; Barbosa et al., 2004), limit the possibility of significant regional tilting after remanence acquisition.

The São Francisco-Congo craton poles are shown in the Gondwana configuration (rotation pole 45.5°N, 32.2°W, 58.2°; Lowver and Scotese, 1987) (Fig. 17a). All the poles fall close to each other, at the northeastern and northern part of Africa, and the Jequié pole is the best dated record. Dubious age constraints for other poles preclude recognition of a firm Paleoproterozoic APW path for the craton. Below we summarize the Paleoproterozoic paleomagnetic record for Precambrian shields surrounding the Congo-São Francisco in the original Gondwana configuration, and compare them to the Jequié pole. Precambrian shields discussed are: Kaapvaal, Guiana, West Africa and Rio de la Plata.

The Kaapvaal block, in South Africa, presents the best paleomagnetic database. Its Paleoproterozoic APW path (Fig. 17b) was recently discussed by de Kock et al. (2006). According to these authors the Lower Swaershoek Formation (WUBS-I) and the Phalabora (PB1) poles situated in northwestern Africa (Fig. 17b) define the 2.05–2.06 Ga position of Kaapvaal craton. Later on, the Kaapvaal APW path drifts southwest along the well dated 2023 ± 4Ma Vredefort pole (VRED, Salminen et al., 2009), the Bushveld Complex poles, and the Limpopo metamorphic "A" pole (Morgan, 1985), whose age is poorly constrained (2.02–1.97 Ga; Table 3). An almost stationary position for the Kaapvaal craton was suggested by Salminen et al. (2009) because of the close position of the 2.05 Ga (WUBS-I) and 2.023 Ga (VRED) poles. The upper Swaershoek/Alma Formations pole (WUBS-II), in spite of its poorly constrained age (1.93–2.05 Ga), defines the southwesterly shift of the Kaapvaal APW path to younger ages. Then a loop is suggested passing through the ~1.93 Ga Hartley lava (HAR) pole, the poorly dated poles of Mashonaland sills (MASH) and Mamatwan (MAM-2), ending with

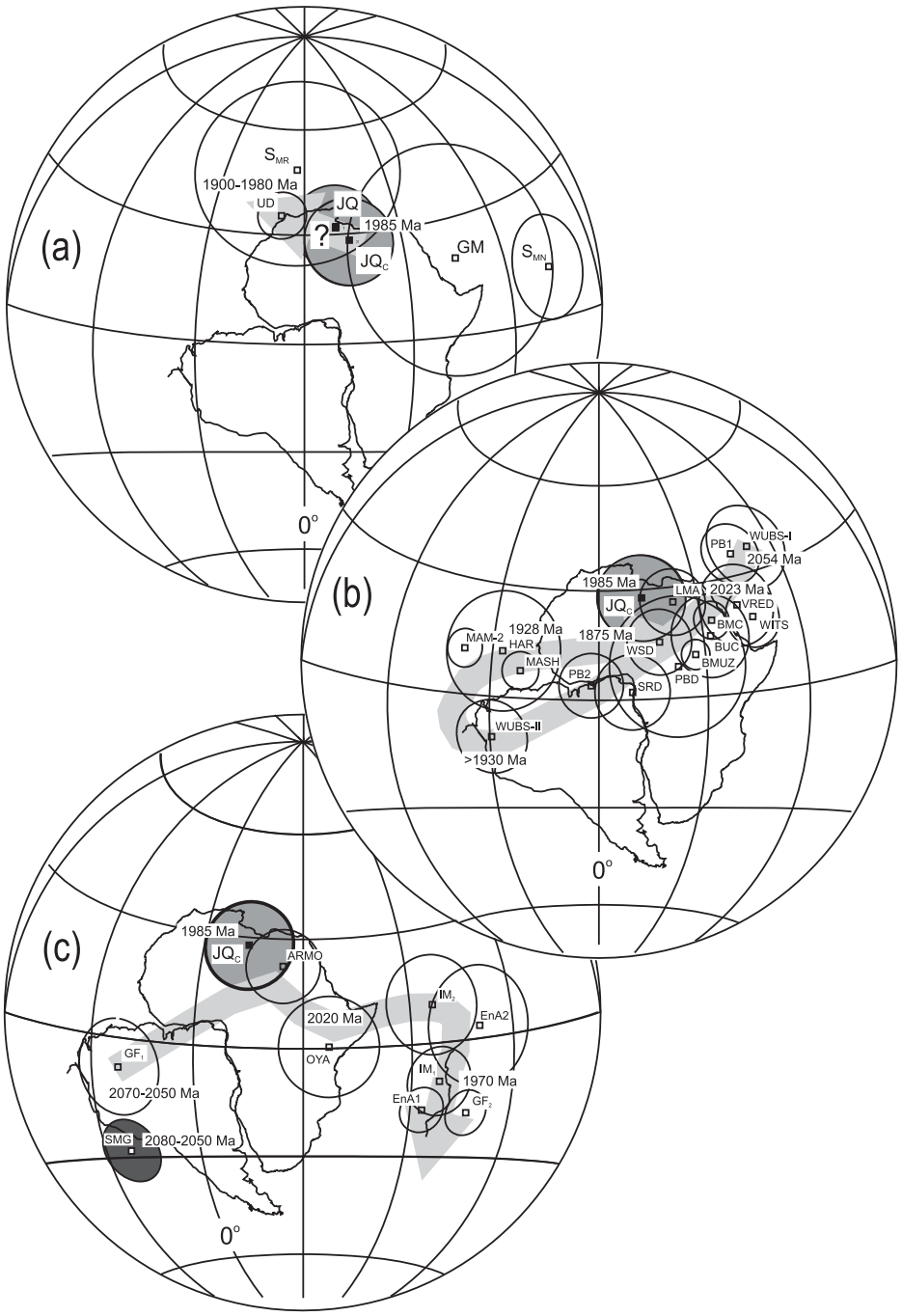


Fig. 17. (a) Paleomagnetic poles from Congo-São Francisco craton. The Jequié pole is shown before (JQ) and after AMS correction (JQ_C); (b) The Jequié pole (JQ_C) is compared with the APW path constructed for the Kaapvaal craton between 2050 Ma and 1870 Ma (after de Kock et al., 2006; Salminen et al., 2009). (c) The Jequié pole (JQ_C) is compared with the APW path constructed for the Amazonian craton between 2060 Ma and 1970 Ma (after Théveniaut et al., 2006). The pole SMG obtained for the Rio de la Plata craton is also shown. Poles (Table 3) rotated to the Gondwana configuration (Africa in its present position).

the well dated 1.88 Ga WSD pole (Fig. 17b). After rotation to a pre-Atlantic, Gondwana configuration, the Jequié pole (JQc) compares well with the Bushveld and Limpopo metamorphic poles and is slightly to the west of the 2.023 Ga Vredefort pole (VRED). As previously stated, the Jequié pole is younger, ca. 1985 Ma. So, unless the suggested stationary position for the Kaapvaal (Salminen et al., 2009) lasted up to 1.99 Ga ago, these independent landmasses probably did not belong to the same plate at that time.

Selected Paleoproterozoic poles (2.15–1.97 Ga) from the Amazonian craton (Guiana Shield) are presented in Table 3. From the first results, obtained by Onstott and Hargraves (1981) and Onstott et al. (1984), a connection with the West Africa shield was proposed after rotation of Guiana poles to the Gondwana configuration. This rotation aligns major, continental-scale mylonitic zones from both shields: the Guri fault in Guiana and the Sassandra fault in West Africa. After that, Nomade et al. (2003) presented new paleomagnetic poles for the French Guiana (their GUI1, GUI2, OYA poles) and proposed a new APW path for the Amazonian craton between 2.04 and 1.99 Ga. Comparison of this APW path with paleomagnetic data from West Africa led these authors to suggest a docking of Amazonia and West Africa at ca. 2.0 Ga ago in a configuration similar to that of Onstott and Hargraves (1981). More recently, the APW path of the Amazonian craton was reviewed by Théveniaut et al. (2006). These authors included new paleomag-

netic and geochronological data from plutonic and metamorphic rocks from French Guiana and re-evaluated the age of previously published poles privileging the Ar–Ar ages on amphiboles, which present closure temperatures similar to the unblocking temperatures of magnetite. This APW path is reproduced in Fig. 17c for the interval between 2.07 and 1.97 Ga after rotation of poles to Gondwana configuration. According to this APW path, the group of poles falling in northern South America (GF_1 combined pole, Table 3) is coeval to the Orosirian deformation event at 2.07–2.05 Ga. After that, the APW path shifts to the east (over the northeastern Africa) following the ARMO and OYA poles (Table 3). An age of 2020 ± 4 Ma is attributed to the OYA pole based on its Ar-Ar amphibole determination (Nomade et al., 2001). The youngest (Paleoproterozoic) Amazonia APW path is defined by a group of poles composed by the Imataca Complex (Onstott and Hargraves, 1981), La Encrucijada granites (Onstott et al., 1984) and four poles for rocks from northern French Guiana (Théveniaut et al., 2006), whose mean was calculated here (GF_2 pole, Table 3). Théveniaut et al. (2006) established the age of this part of the APW path based on the ^{40}Ar–^{39}Ar age of 1972 ± 4 Ma (amphibole) for the La Encrucijada intrusive granites (Onstott et al., 1984). The cooling history of the Imataca Complex based on hornblende, biotite, and feldspar dating suggests a similar age (ca. 1.97 Ga) for the Imataca pole (Onstott et al., 1989). In Fig. 17c, after rotation to the Gondwana configuration, the Jequié pole (JQc) is compared with the Amazonian craton APW path. The JQc pole compares well with the ARMO pole (ca. 2030 Ma) from Théveniaut et al. (2006) and is slightly to the northwest of the 2.02 Ga OYA pole. Here again, the age assigned to the Jequié pole is younger (ca. 1.99 Ga), thus precluding the connection between Congo-São Francisco and proto-Amazonian craton at Early Proterozoic times.

Finally, the Rio de la Plata Craton, which is located at the southwestern border of Congo-São Francisco in Atlantica presents only a single published Paleoproterozoic pole (SMG) (Table 3). This pole was obtained for the Soca and Isla Mala granites, thought to be part of the Piedra Alta Terrane. U–Pb SHRIMP dating of 2056 ± 6 Ma for Soca granite, and 2065 ± 9 and 2074 ± 6 Ma for Isla Mala granite suggests an age between 2050 and 2080 Ma for this pole (Badgen et al., 2009). After rotation to the Gondwana configuration, the SMG pole is compared with the Guiana Shield APW path in Fig. 17c. Even though the SMG pole falls close to the GF1 pole of similar age, the angular difference between these poles, not overlapping within error (α_{95}), suggests that these blocks were separate, though perhaps not distant, at ca. 2.06 Ga ago.

5.3. Atlantica, Ur and beyond

The Paleoproterozoic dynamics is marked by the development of numerous large orogenic belts (intra-oceanic and continental arcs), which could have formed because of the Columbia supercontinent assembly (Rogers and Santosh, 2002). According to these authors, Columbia was formed by three large blocks known as Atlantica, Ur and Nena (Rogers, 1996), which are thought to have remained intact during the formation of the later Rodinia and Pangea Supercontinents. In the new models of Columbia (e.g., Zhao et al., 2002, 2004; Kusky et al., 2007; Hou et al., 2008; Bispo-Santos et al., 2008), which included other formerly forgotten small blocks, Atlantica and Nena are mostly unchanged. Ur, however, was dismembered with South Africa attached to Australia and India still linked to North China (Zhao et al., 2002, 2004; Hou et al., 2008). Kusky et al. (2007) maintain the Ur block and place North China between Baltica and Atlantica, based on the continuity of Paleoproterozoic belts between North China (North Obei mobile belt) and the proto-Amazonian craton (1.98–1.80 Ga Ventuari-Tapajos province; Cordani and Teixeira, 2007). Bispo-Santos et al. (2008) follow the same reasoning but rotates North China ca. 90° based on the paleomagnetic data. In their reconstruction the Paleoproterozoic Trans-North China mobile belt (Zhao et al., 2005, 2006) in North China represents the continuity of the Ventuari-Tapajos belt (proto-Amazonian craton).

The new Jequié pole can be used to test the entity of Atlantica, and its relation with Ur in the Columbia model of Rogers and Santosh (2002). Comparison of the Jequié pole with poles from Kaapvaal, proto-Amazonian craton and West Africa suggests that these units were not part of the same supercontinent at ca. 1.99 Ga. If we accept an age between 1900 and 1980 Ma for the Uauá pole, we can trace a short APW path for the São Francisco/Congo craton (Fig. 17a). This APW path is different from the traced Amazonian craton APW path (Fig. 17c) for the same time interval suggesting that Atlantica if it really existed, it was short lived. This is further substantiated by robust geological and paleomagnetic evidence indicating the existence of a wide Clymene ocean separating the Amazon-West Africa craton from the rest of Gondwana (Trindade et al., 2006; Tohver et al., 2010). On the western side of the SFC, Pimentel et al. (1999, 2000) support the presence of juvenile magmatism dated at 930 Ma and 830 Ma in central Brazil, with metamorphic ages as old as 760 Ma at the western margin of the Central Goiás Complex indicating an early collisional episode on the western margin of the craton (Pimentel et al., 2000). The presence of large oceans between São Francisco and Amazonian cratons since at least 800 Ma ago is later emphasized by several authors based on geological and paleomagnetic grounds (Kröner and Cordani, 2003; Cordani et al., 2003a,b, 2009; D'Agrella-Filho et al., 2004; Trindade et al., 2006; Tohver et al., 2006, 2010).

6. Summary and conclusions

A new paleomagnetic pole at 339.6°E, 5.4°N (A_{95} = 11.2°) was obtained for 12 sites (near Mutuípe city) of high grade metamorphic rocks from the Archaean/Paleoproterozoic Jequié block, SFC. After AMS correction a new statistically improved paleomagnetic pole at 342.1°E, −0.5°N (N = 12, A_{95} = 9.6°) was calculated. ^{40}Ar–^{39}Ar plateau ages of 2035 ± 4 Ma (hornblende) and 1876 ± 5 Ma (biotite) were obtained for sample from Mutuípe region suggesting a long time of exhumation and cooling, succeeding the tectonic stability of the marginal Itabuna–Salvador–Curaçá belt. Considering the cooling rate of 1.4 °C/Ma between amphibole and biotite ages, the age of magnetization would be comprised between 1985 and 2089 Ma, according to Pullaiah et al.'s (1975) equation. This pole was compared with the present paleomagnetic poles from South America and Africa to test the Atlantica continental block of Rogers (1996) and the Paleoproterozoic Columbia supercontinent (Rogers and Santosh, 2002). The paleomagnetic data suggest that if Columbia existed as envisaged by these authors it would be short lived. The same can be said for Atlantica but Neoproterozoic geological and paleomagnetic evidence show that this continental block most probably did not exist.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.precamres.2011.01.008.

References

Alkmim, F.F., 2004. O que faz de um cráton um cráton? O Cráton do São Francisco e as revelações almeidianas ao delimitá-lo. In: Mantesso-Neto, V., Bartorelli, A., Carneiro, C.D.R., Brito Neves, B.B. de. (Org.), Geologia do Continente Sul Americano, Evolução da obra de Fernando Marques de Almeida, São Paulo, Brazil Beca, pp. 17–35.

Badgen, E., Rapalini, A.E., Sanchez-Bettucci, L., Vazquez, C.A., 2009. The first Paleoproterozoic (ca. 2.05 Ga) paleomagnetic pole for the Piedra Alta Terrane (Rio de la Plata craton), Uruguay. In: Primera Reunión Bienal de LATINMAG (CDROM), Isla de Margarita, Venezuela, 2009.

Barbosa, J.S.F., 1990. The granulites of the Jequié Complex and the Atlantic coast mobile belt, southern Bahia, Brazil-an expression of Archaean/early Proterozoic plate convergence. In: Vielzeuf, D., Vidal, Ph. (Eds.), Granulites and Crustal Evolution. Kluwer Academic Plublishers, Dordrecht, pp. 195–221.

Barbosa, J.S.F., Sabaté, P., 2002. Geological features and the Paleoproterozoic collision of four Archean crustal segments of the São Francisco Craton, Bahia, Brazil. A syntesis. Anais da Acad. Bras. de Ciên. 74 (2), 343–359.

Barbosa, J.S.F., Sabaté, P., 2003. Colagem Paleoproterozóica de placas arqueanas do Cráton do São Francisco na Bahia. Rev. Bras. Geoc. 33, 7–14.

Barbosa, J.S.F., Sabaté, P., 2004. Archaean and Palaeoproterozoic crustal of the São Francisco Craton, Bahia, Brazil: geodynamic features. Precambrian Res. 133, 1–27.

Barbosa, J.S.F., Correa-Gomes, L.C., Marinho, M.M., Silva, F.C.A., 2003. Geologia do segmento sul do orógeno Itabuna–Salvador–Curaçá. Rev. Bras. de Geoc. 33 (1-Suplemento), 33–48.

Barbosa, J.S.F., Martin, H., Peucat, J.J., 2004. Paleoproterozoic dome forming structures related to granulite facies metamorphism. Jequié Block, Bahia, Brasil: petrogenetic approaches. Precambrian Res. 135, 105–131.

Barbosa, J.S.F., Peucat, J.J., Martin, H., Silva, F.A., Moraes, A.M., Corrêa-Gomes, L.C., Sabaté, P., Marinho, M.M., Fanning, C.M., 2008. Petrogenesis of the late-orogenic Bravo granite and surrounding high-grade country rocks in the Palaeoproterozoic orogen of Itabuna–Salvador–Curaçá block, Bahia, Brazil. Precambrian Res. 167, 35–52.

Bastos Leal, L.R., Teixeira, W., Piccirillo, E.M., Meneses Leal, A.B., Girardi, V.A.V., 1994. Geocronologia Rb/Sr e K/Ar do enxame de diques máficos de Uauá, Bahia (Brasil). Geochim. Brasiliensis 8, 99–114.

Bates, M.P., Jones, D.L., 1996. A palaeomagnetic investigation of the Mashonaland dolerites, north-east Zimbabwe. Geophys. J. Int. 126, 513–524.

Bispo-Santos, F., D'Agrella-Filho, M.S., Pacca, I.I.G., Janikian, L., Trindade, R.I.F., Elming, S.-Å., Silva, J.A., Barros, M.A.S., Pinho, F.E.C., 2008. Columbia revisited: paleomagnetic results from the 1790 Ma Colíder volcanics (SW Amazonian Craton, Brazil). Precambrian Res. 164, 40–49, doi:10.1016/j.precamres.2008.03.004.

Brito Neves, B.B., de Cordani, U.G., Torquato, J.R.F., 1980. Evolução geocronológica do Precambriano do Estado da Bahia. In: Inda, H.A.V., Duarte, F.B. (Eds.), Geologia e Recursos Minerais do Estado da Bahia, Textos Básicos, Secretaria de Minas e Energia/Coordenação de Produção Mineral (1980), Salvador, 3, p. 123.

Buick, I.S., Hermann, J., Williams, I.S., Gibson, R.L., Rubatto, D., 2006. A SHRIMP U–Pb and LA-ICP-MS trace element study of the petrogenesis of garnet-cordierite-orthoamphibole gneisses from the Central Zone of the Limpopo Belt, South Africa. Lithos 88, 150–172.

Constable, C., Tauxe, L., 1990. The bootstrap for magnetic susceptibility tensor. J. Geophys. Res. 95, 8383–8395.

Cordani, U.G., Iyer, S.S., 1978. Geochronological investigation on the Precambrian granulitic terrain of Bahia-Brazil. Precanbrian Res. 9, 255–274.

Cordani, U.G., Teixeira, W., 2007. Proterozoic accretionary belts in the Amazonian Craton. In: Hatcher, R.D. Jr., Carlson, M. P., McBride, J. H., and Martinez Catalán, J. R. (Org.). The 4-D Framework of Continental Crust. GSA Memoir, Boulder, Colorado, Geological Society of America Book Editors, 2007, v. 200, pp. 297–320.

Cordani, U.G., Brito-Neves, B.B., D'Agrella-Filho, M.S., 2003a. From Rodinia to Gondwana: a review of the available evidence from South America. Gondwana Res. 6, 275–283.

Cordani, U.G., D'Agrella-Filho, M.S., Brito-Neves, B.B., Trindade, R.I.F., 2003b. Tearing up Rodinia: the Neoproterozoic paleogeography of South American fragments. Terra Nova 15, 350–359, doi:10.1046/j.1365-3121.2003.00506.x.

Cordani, U.G., Iyer, S.S., Taylor, P.N., Kawashita, K., Sato, K., McReath, I., 1992. Pb–Pb, Rb–Sr and K–Ar systematics of the Lagoa Real Uranium Province (South-central Bahia, Brazil) and the Espinhaço Cycle (ca. 1.5–1.0 Ga). J. South Am. Earth Sci. 5, 33–46.

Cordani, U.G., Sato, K., Marinho, M.M., 1985. The geologic evolution of the ancient granite-greenstone terrane of central-southern Bahia, Brazil. Precambrian Res. 27, 187–213.

Cordani, U.G., Teixeira, W., D'Agrella-Filho, M.S., Trindade, R.I., 2009. The position of the Amazonian Craton in supercontinents. Gondwana Res. 15, 396–407.

D'Agrella-Filho, M.S., Pacca, I.I.G., 1998. Paleomagnetism of Paleoproterozoic mafic dyke swarm from the Uauá region, northeastern São Francisco Craton, Brazil: tectonic implications. J. South Am. Earth Sci. 11, 23–33.

D'Agrella-Filho, M.S., Feybesse, J.L., Prian, J.P., Dupuis, D., N'Dong, 1996. Palaeomagnetism of Precambrian rocks from Gabon, Congo craton, Africa. J. Afr. Earth Sci. 22, 65–80.

D'Agrella-Filho, M.S., Pacca, I.G., Trindade, R.I.F., Teixeira, W., Raposo, M.I.B., Onstott, T.C., 2004. Paleomagnetism and $^{40}Ar/^{39}Ar$ ages of mafic dykes from Salvador (Brazil): new constraints on the São Francisco Craton APW path between 1080 and 1010 Ma. Precambrian Res. 132, 55–77, doi:10.1016/j.precamres.2004.02.003.

Dahl, P.S., 1996. The effects of composition on retentivity of argon and oxygen in hornblende and related amphiboles: a field-tested empirical model. Geochim. Cosmochim. Acta 60, 3687–3700.

Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence. Phys. Earth Planet. Interiors 13, 260–267.

de Kock, M.O., Evans, D.A.D., Dorland, H.C., Beukes, N.J., Gutzmer, J., 2006. Paleomagnetism of the lower unconformity-bounded sequences of the Waterberg Group, South Africa: towards a better-defined apparent polar wander path for the Paleoproterozoic Kaapvaal Craton. South Afr. J. Geol. 109, 157–182.

Dunlop, D.J., 2002a. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc) 1. Theoretical curves and tests using titanomagnetite data. J. Geophys. Res. 107 (B3), 4-1–4-22, doi:10.1029/2001JB000486.

Dunlop, D.J., 2002b. Theory and application of the Day plat (Mrs/Ms versus Hcr/Hc) 2. Application to data for rocks, sediments, and soils. J. Geophys. Res. 107 (B3), 5-1–5-15, doi:10.1029/2001JB000487.

Evans, D.A.D., Gutzmer, J., Beukes, N.J., Kirschivink, J.I., 2001. Paleomagnetic constraints on ages of mineralization in the Kalahari Manganesi Field, South Africa. Econ. Geol. 96, 621–631.

Evans, D.A.D., Beukes, N.J., Kirschvink, J.L., 2002. Paleomagnetism of a lateritic paleoweathering horizon and overlying Paleoproterozoic red beds from South Africa: implications for the Kaapvaal apparent polar wander path and confirmation of atmospheric oxygen enrichment. J. Geophys. Res. 107 (B12), 2-1–2-22, doi:10.1029/2001JB000432, 2326.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A217, 295–305.

French, J.E., Heaman, L.M., Chacko, T., 2002. Feasibility of chemical U-Th-total PB baddeleyite dating by electron microprobe. Chem. Geol. 188, 85–104.

Frost, B.R., 1991. Magnetic petrology: factors that control the occurrence of magnetite in crustal rocks. In: Lindsley, D.H. (Ed.), Oxide Minerals: Petrologic and Magnetic Significance, Reviews in Mineralogy, vol. 25. , pp. 489–509.

Ghiorso, M.S., 1997. Thermodynamic analysis of the effect of magnetic ordering on miscibility gaps in the Fe-Ti cubic and hombohedral oxide minerals and the Fe-Ti oxide geothermometer. Phys. Chem. Miner. 25, 28–38.

Hanson, R.E., Gose, W.A., Crowley, J.L., Ramezani, J., Bowring, S.A., Bullen, D.S., Hall, R.P., Pancake, J.A., Mukwakwami, J., 2004. Paleoproterozoic intraplate magmatism and basin development on the Kaapvaal craton: age, paleomagnetism and geochemistry of ~1. 93 to 1. 87 Ga post Waterberg dolerites. South Afr. J. Geol. 107, 233–254.

Hattingh, P.J., 1989. Palaeomagnetism of the upper zone of the Bushveld Complex. Tectonophysics 165, 131–142.

Hattingh, P.J., 1999. Palaeomagnetism and the Bushveld Complex in Southern Africa: some enigmas. Memoir Geol. Soc. India 44, 1–7.

Hou, G., Santosh, M., Qian, X., Lister, G.S., Li, J., 2008. Configuration of the Late Paleoproterozoic supercontinent Columbia: insights from radiating mafic dyke swarms. Gondwana Res. 14, 395–409, doi:10.1016/j.gr.2008.01.010.

Jackson, M., 1991. Anisotropy of magnetic remanence: a brief review of mineralogical sources, physical origins and geological applications, and comparison with susceptibility anisotropy. Pure Appl. Geophys. 136, 1–28.

Jelinek, V., 1981. Characterization of the magnetic fabric of rocks. Tectonophysics 79, 63–67.

Kirshvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astro. Soc. 62, 699–718.

Kröner, A., Cordani, U.G., 2003. African and South American cratons were not part of the Rodinia supercontinent: evidence from field relationships and geochronology. Tectonophysics 375, 325–352.

Kusky, T., Li, J., Santosh, M., 2007. The paleoproterozoic North Hebei Orogen: North China craton's collisional suture with the Columbia supercontinent. Gondwana Res. 12, 4–28, doi:10.1016/J.gr.2006.11.012.

Layer, P.W., Kröner, A., McWilliams, M., Burghele, A., 1988. Paleomagnetism and age of the Archean Usushwana Complex, Southern Africa. J. Geophys. Res. 93, 449–457.

Ledru, P., Cocherie, A., Barbosa, J., Johan, V., Onstott, T., 1994. Ages du métamorphisme granulitique dans le craton du São Francisco (Brésil). Implications sur la nature de l'orogène transamazonien. C. R. Acad. Sci. Paris t. 318 (série II), 251–257.

Letts, S., Torsvik, T.H., Webb, S.J., Ashwal, L.D., Eide, E.A., Chunnett, G., 2005. Palaeomagnetism and $^{40}Ar/^{39}Ar$ geochronology of mafic dykes from the eastern Bushveld Complex (South Africa). Geophys. J. Int. 162, 36–48.

Lowver, L., Scotese, C.R., 1987. A revised reconstruction of Gondwanaland. In: McKenzie, G.D. (Ed.), Gondwana Six: Structure, Tectonics and Geophysics, Am. Geophys. Un. Mon., vol. 40, pp. 17–23.

Marinho, M.M., Vidal, Ph., Alibert, C., Barbosa, J.S.F., Sabaté, P., 1994. Geochronology of the Jequié-Itabuna granulitic belt and of the Contendas-Mirante volcano-sedimentary belt. Boletim do Instituto de Geociências, Universidade de São Paulo, Brazil, Publicação Especial 17, pp. 73–96.

McDougall, I., Harrison, M., 1999. Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method. Oxford University Press, 269 pp.

McElhinny, M.W., Opdyke, N.D., 1964. The paleomagnetism of the Precambrian dolerites of Eastern Southern Rhodesia, an example of geologic correlation by rock magnetism. J. Geophys, Res. 69, 2465–2475.

McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astro. Soc. 67, 19–33.

McFadden, P., McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophys. J. Int. 103, 725–729.
Morgan, G.E., 1985. The paleomagnetism and cooling hystory of metamorphic and igneous rocks from the Limpopo Mobile Belt, Southern Africa. Geol. Soc. Am. Bull. 96, 663–675.
Morgan, G.E., Briden, J.C., 1981. Aspects of Precambrian palaeomagnetism, with new data from the Limpopo mobile belt and Kaapvaal Craton in Southern Africa. Phys. Earth Planet. Interiors 24, 142–168.
Nomade, S., Chen, Y., Féraud, G., Pouclet, A., Théveniaut, H., 2001. First paleomagnetic and $^{40}Ar/^{39}Ar$ study of Paleoproterozoic rocks from the French Guyana (Camopi and Oyapok rivers), northeastern Guyana Shield. Precambrian Res. 109, 239–256.
Nomade, S., Chen, Y., Pouclet, A., Féraud, G., Théveniaut, H., Daouda, B.Y., Vidal, M., Rigolet, C., 2003. The Guiana and the West African Shield Palaeoproterozoic grouping: new palaeomagnetic data for French Guiana and the Ivory Coast. Geophys. J. Int. 154, 677–694.
Oliveira, E.P., Windley, B.F., McNaughton, N.J., Pimentel, M., Fletcher, I.R., 2004. Contrasting copper and chromium metallogenic evolution of terranes in the Palaeoproterozoic Itabuna–Salvador–Curacá orogen, São Francisco craton, Brazil: new zircon (SHRIMP) and Sm–Nd (model) ages and their significance for orogen-parallel escape tectonics. Precambrian Res. 128, 143–165.
Onstott, T.C., Hall, C.M., York, D., 1989. $^{40}Ar/^{39}Ar$ thermochronometry of the Imataca Complex, Venezuela. Precambrian Res. 42, 255–291.
Onstott, T.C., Hargraves, R.B., 1981. Proterozoic transcurrent tectonics: palaeomagnetic evidence from Venezuela and Africa. Nature 289, 131–136.
Onstott, T.C., Hargraves, R.B., York, D., Hall, C., 1984. Constraints on the motions of South American and African shields during the Proterozoic: I. $^{40}Ar/^{39}Ar$ and paleomagnetic correlations between Venezuela and Liberia. Geol. Soc. Am. Bull. 95, 1045–1054.
Pimentel, M.M., Fuck, R.A., Botelho, N.F., 1999. Granites and the geodynamic history of the neoproterozoic Brasília belt, Central Brazil: a review. Lithos 46, 463–483.
Pimentel, M.M., Fuck, R.A., Jost, H., Ferreira-Filho, C.F., Araújo, S.M., 2000. The basement of the Brasília Fold Belt and the Goiás Magmatic arc. In: Cordani, U.G., Milani, E.J., Thomaz-Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America, Rio de Janeiro. , pp. 195–229.
Pullaiah, G., Irving, E., Buchan, K.L., Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift, Earth and Planet. Sci. Lett. 28, 133–143.
Raposo, M.I.B., D'Agrella-Filho, M.S., Siqueira, R., 2003. The effect of magnetic anisotropy on paleomagnetic directions in high-grade metamorphic rocks from the Juiz de Fora Complex. SE Brazil. Earth Planet. Sci. Lett. 209, 131–147, doi:10.1016/S0012-821X(03)00065-7.
Reischmann, T., 1995. Precise U–Pb age determination with baddeleyite (ZrO_2), a case study from the Phalaborwa Igneous Complex, South Africa. South Afr. J. Geol. 98, 1–4.
Rochette, P., Jackson, M., Aubourg, C., 1992. Rock magnetism and the interpretation of anisotropy magnetic susceptibility. Rev. Geophys. 30, 209–226.
Rosa, M.L.S., Conceicão, H., Macambira, M.J.B., Scheller, T., Martin, H., Bastos-Leal, L.R., 2001. Idades Pb-Pb e assinatura isotópica Rb-Sr e Sm- Nd do magmatismo sienítico paleoproterozóico no sul do Cinturão Móvel Salvador-Curaçá: Maciço Sienítico de São Felix. Bahia. Rev. Bras. Geoc. 31 (3), 390–400.
Rogers, J.J.W., 1996. A history of continents in the past three billion years. J. Geol. 104, 91–107.
Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic Supercontinent. Gondwana Res. 5, 5–22.
Salminen, J., Pesonen, L.J., Reimold, W.U., Donadini, F., Gibson, R.L., 2009. Paleomagnetic and rock magnetic study of the Vredfort impact structure and the Johannesburg Dome, Kaapvaal Craton, South Africa-Implications for the apparent polar wander path of the Kaapvaal Craton during the Mesoproterozoic. Precambrian Res. 168, 167–184, doi:10.1016/j.precamres.2008.09.005.
Samson, S.D., Alexander, E.C., 1987. Calibration of the interlaboratory $^{40}Ar/^{39}$ standard Mmhb-1. Chem. Geol. 66, 27–34.
Silva, L.C., Armstrong, R., Delgado, I.M., Pimentel, M., Arcanjo, J.B., Melo, R.C., Teixeira, L.R., Jost, H., Carcoso Filho, J.M., Pereira, L.H.M., 2002. Reavaliação da evolução geológica em terrenos Précambrianos brasileiros com base em novos dados U–Pb SHRIMP. Parte I: Limite centro-oriental do Cráton do São Francisco. Rev. Bras. Geoc. 33 (4), 501–512.
Teixeira, W., Sabaté, J.P., Barbosa, J., Noce, C.M., Carneiro, A.M., 2000. Archean and Paleoproterozoi tectonic evolution of the São Francisco Craton. In: Cordani, U., Thomaz Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America, 31st Intern. Geol. Congr., Rio de Janeiro. , pp. 101–138.
Théveniaut, H., Delor, C., Lafon, J.M., Monié, P., Rossi, P., Lahondère, D., 2006. Paleoproterozoic (2155–1970 Ma) evolution of the Guiana Shield (Transamazonian event) in the light of new paleomagnetic data from French Guiana. Precambrian Res. 150, 221–256, doi:10.1016/j.precamres.2006.08.004.
Tohver, E., D'Agrella-Filho, M.S., Trindade, I.F., 2006. Paleomagnetic record of Africa and South America for the 1200–500 Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambrian Res. 147, 193–222, doi:10.1016/j.precamres.2006.01.015.
Tohver, E., Trindade, R.I.F., Solum, J.G., Hall, C.M., Riccomini, C., Nogueira, A.C., 2010. Closing the Clymene Ocean and Bending a Brasiliano belt: Evidence for the Cambrian formation of Gondwana from SE Amazon craton. Geology 38, 267–270, doi:10.1130/G30510.1.
Trindade, R.I.F., D'Agrella-Filho, M.S., Epof, I., de Brito-Neves, B.B., 2006. Paleomagnetism of the Early Cambrian Itabaiana mafic dikes, NE Brazil, and implications for the final assembly of Gondwana and its proximity to Laurentia. Earth Planet. Sci. Lett. 244, 361–377, doi:10.1016/j.epsl.2005.12.039.
Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.
Zhao, G., Cawood, P.A., Wilde, S.A., Sun, M., 2002. Review of global 2.1–1.8 Ga orogens: implications for a pre-Rodinia supercontinent. Earth Sci. Rev. 59, 125–162.
Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth Sci. Rev. 67, 91–123.
Zhao, G., Sun, M., Wilde, S.M., Sanzhong, L., 2005. Late Archean to Paleoproterozoic evolution of the North China Craton: key issues revisited. Precambrian Res. 136, 177–202, doi:10.1016/j.precamres.2004.10.002.
Zhao, G., Sun, M., Wilde, S.M., Sanzhong, L., Zhang, J., 2006. Some key issues in reconstructions of Proterozoic supercontinents. J. Asian Earth Sci. 28, 3–19, doi:10.1016/j.jseaes.2004.06.010.
Zijderveld, J.D.A., 1967. A. C. Demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.